Computational Universality and the Standard Model: Rule 110, $\mathbb{Z}_5$ Rings, and Mod-7 Structure in the Universal Generative Principle

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Abstract

The Standard Model's generation orbit uniquely and algebraically determines Rule 110 — one of fewer than five computationally universal rules among all 256 binary cellular automata — as the sole binary CA rule consistent with SM winding structure and vacuum-transparency. This derivation is not a coincidence: the orbit directly specifies all seven active Rule 110 minterms, making universality a structural *necessity* of the generation sequence, not an incidental property of the substrate. Independently, the UGP's canonical substrate (the UWCA, proven in [5]) also implements Rule 110. The two derivations meet at the same rule: Rule 110 is forced both *from above* (UGP substrate universality) and *from below* (SM orbit constraints). The GTE mass cascade sector is simultaneously realized on the same Rule 110 tape (tape-level Hypothesis B, $\mathbf{A}_{\text{Lean}}$, zero bare `sorry`). Together, these results give the first mathematically derived, machine-certified identification of which cellular automaton governs specific Standard Model sectors, and why it is uniquely forced in both directions.

The central result (**CUP-4**, $\mathbf{A}_{\text{Lean}}$): the SM generation orbit ($\text{gen}_1 \to \text{gen}_2 \to \text{gen}_3$ under the total-parity morphism φ) algebraically determines all 8 bits of Rule 110 by activating exactly 7 non-trivial binary neighborhoods and specifying their required outputs. Uniqueness is algebraically necessary, not merely combinatorial. Structural p-value $\approx 0.003\%$. The orbit–universality connection is structural: perturbing any orbit bit destroys orbit satisfiability for almost all rules, and the Rule 110 minterm set $\{1, 2, 3, 5, 6\}$ required by the orbit is precisely what enables its glider dynamics and universality.

Three further structural results follow. **CUP-8/9** ($\mathbf{A}_{\text{Lean}}$): every generation-3 SM particle has odd parity; the five SM families form a closed $\mathbb{Z}_5$ ring with 10 equivalent orderings. **CUP-11/12** ($\mathbf{A}_{\text{Lean}}$/**A/D**): a mod-7 CA f_{MDL} satisfying the SM orbit and Rule 110 binary sublayer exists and is computationally universal (CUP-11c, $\mathbf{A}_{\text{Lean}}$, zero `sorry`); its MDL-minimal completion has 76-bit description length and is Lean-uniquely identified. The unique $\mathbb{Z}_7$ cross-dimensional coupling is $\mathbb{Z}_7$ addition ($\mathbf{A}_{\text{Lean}}$). **Garden of Eden** ($\mathbf{A}_{\text{Lean}}$): $\text{gen}_1 = [1, 5, 2, 2, 1]$ has zero predecessors under f_{MDL} across all $7^5 = 16{,}807$ $\mathbb{Z}_7$ states (`native_decide`, zero `sorry`); first-generation stability is arithmetic unreachability.

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Physical incompleteness ($\mathbf{A}_{\text{Lean}}$ conditional): because f_{MDL} is Turing-complete, the vacuum-reachability predicate is halting-undecidable (`fmdl_vacuum_reachability_is_undecidable`, conditional on six bridge axioms).

Tape-level unification ($\mathbf{A}_{\text{Lean}}$): a single Rule 110 Bool tape simultaneously computes both UGP sectors. `hypothesis_b_tape_level` certifies that the SM charge sector (fMDL winding) and the mass cascade (GTE arithmetic) cohere on one infinite tape, with the GTE embedding proved via one named Cook universality axiom. PSC forces this substrate: the chain PSC $\to$ SM gauge $\to$ orbit $\to$ Rule 110 $\to$ dual-sector tape is fully Lean-certified (`psc_forces_tape_unification`, $\mathbf{A}_{\text{Lean}}$).

The complete formalization is zero bare `sorry`; three named physics bridge axioms remain (`rcc_physics_ax`, `rule110_simulates_computable`, `simultaneous_dual_tape_ax`), each backed by substantial partial certification.

1 Introduction

The Universal Generative Principle (UGP) derives the Standard Model particle spectrum from three axioms of locality, symmetry, and compression applied to a number-theoretic sieve at ridge level $n = 10$ [4]. A separate result, proved in [5], establishes that the UGP is *computationally universal*: the Universal Windowed Cellular Automaton (UWCA) arising from the UGP's operational mechanics bisimulates Rule 110, a Turing-complete one-dimensional cellular automaton [1,2].

Until this paper, these two results were disconnected. The computational universality was an observation about what the UGP substrate *can do*, with no known connection to why the Standard Model has the structure it does. This paper closes that gap.

Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A}_{\text{MDL}} > \mathbf{A/D} > \mathbf{B} > \mathbf{D}$$

$\mathbf{A}_{\text{Lean}}$**:** machine-checked exact arithmetic in Lean 4, zero `sorry`.
$\mathbf{A}_{\text{MDL}}$**:** MDL-unique within a declared expression class.
A/D: computationally confirmed; a physics bridge remains.
B: calibrated reproduction.
D: exploratory / frontier.

Table 1: Reviewer's map: what is being claimed, at what strength, and where to verify.

Result	Status	Claim	Where
CUP-4 uniqueness	$\mathbf{A}_{\text{Lean}}$	SM orbit + vacuum-transparency algebraically determines all 8 Rule 110 outputs. Exhaustive Lean proof over 256 rules.	§2, App. A
CUP-8	$\mathbf{A}_{\text{Lean}}$	All SM generation-3 particles have odd total parity.	§3
CUP-9 ($\mathbb{Z}_5$)	$\mathbf{A}_{\text{Lean}}$	5 SM families form a closed $\mathbb{Z}_5$ ring; 10 equivalent orderings.	§3
$p \approx 0.003\%$	$\mathbf{A}_{\text{Lean}}$	Structural significance of CUP-4 result.	§4, App. B
Orbit–universality	**A**	Orbit directly specifies Rule 110 minterms $\{1, 2, 3, 5, 6\}$; structural, not coincidental.	§5
CUP-11c	$\mathbf{A}_{\text{Lean}}$	Mod-7 universal CA satisfying SM orbit exists (Lean, zero sorry).	§6
CUP-11a/b	**A/D**	Mod-7 CA rule algebraic form and Z_7 sum conservation.	§6
CUP-12	$\mathbf{A}_{\text{Lean}}$	MDL-minimal $\mathbb{Z}_7$ CA (76-bit); f_{MDL} uniquely MDL-minimal (Lean-proved).	§7, §8
Z_7 coupling	$\mathbf{A}_{\text{Lean}}$	Unique P22-consistent Z_7 coupling is Z_7 addition (Lean-proved).	§8
3D spacetime	**A**	$f_{\text{MDL,3D}}$ gives 3+1D causal geometry by lattice construction.	§9
All 5 SM windings	**A**	$Z_7 \in \{0, 2, 3, 4, 6\}$ generated. Electron $Z_7 = 4$ cross-dim only (Lean).	§9
Matter dominance	**A**	Structural Z_7 CP violation from orbit construction.	§9
GoE stability	$\mathbf{A}_{\text{Lean}}$	gen_1 has zero $\mathbb{Z}_7$ predecessors under f_{MDL} (all 7^5 states, `native_decide`, zero `sorry`).	§7, §12.6
Physical incompleteness	$\mathbf{A}_{\text{Lean}}$ (cond.)	Vacuum-reachability undecidable for binary X; Lean zero `sorry` conditional on Cook–Wolfram–APS bridge axioms.	§12
Dark sector hint	**D**	Mirror-dual Z_2 branch as dark sector; arithmetic dark charge.	§10
GTE compilation	$\mathbf{A}_{\text{Lean}}$	T = `uwca_eval` Σ_{GTE} for 1-tile finite set Σ_{GTE}; four-component `hypothesis_a_complete` zero `sorry`.	§12, App. A
Hyp. B (full)	$\mathbf{A}_{\text{Lean}}$	Both sectors certified zero `sorry`: fMDL tape-level coherence (`fmdl_sector_coherence`) and GTE arithmetic-level coherence (`gte_sector_coherence`); existential form `hypothesis_b_single_substrate` packages both.	App. A
Hyp. B (tape level)	$\mathbf{A}_{\text{Lean}}$	`gte_embeds_in_rule110` proves GTE arithmetic embeds in Rule 110 at the infinite-tape level (zero `sorry`, one bridge axiom); `hypothesis_b_tape_level` packages tape-level unification of both sectors.	App. A

Central result

The Standard Model's generation orbit *algebraically forces* Rule 110 — one of fewer than five computationally universal rules among 256 binary CA rules — and this forcing is structural, not coincidental. The orbit directly specifies all seven active Rule 110 minterms (Proposition 2.3, $\mathbf{A}_{\text{Lean}}$). The unique $\mathbb{Z}_7$ extension (f_{MDL}, CUP-12) and the P22 [9] winding-conservation theorem together force the SM vertex rules. The whole chain from orbit to vertex rules is machine-certified in Lean 4 (Figure 1).

The central finding is a causal chain:

$$
\begin{aligned}
&\text{SM orbit constraints}\\
&\quad\Longrightarrow \text{ algebraically determines Rule 110 minterms } \{1,2,3,5,6\}\\
&\quad\Longrightarrow \text{ uniquely selects Rule 110 (CUP-4, Lean-proved)}\\
&\quad\Longrightarrow \text{ Rule 110 is computationally universal (P04 [5], Lean-proved)}\\
&\quad\Longrightarrow \mathbb{Z}_7 \text{ orbit structure of } f_{\mathrm{MDL}} \text{ (CUP-11, Lean-proved)}\\
&\quad\Longrightarrow \text{ unique } \mathbb{Z}_7 \text{ coupling (Theorem 8.1, } \mathbf{A}_{\mathrm{Lean}})\\
&\quad\Longrightarrow \text{ SM winding-conservation vertex rules (P22 [9], Lean-proved).}
\end{aligned}
\tag{1}
$$

The orbit–universality link (§5) is proved to be structural rather than coincidental: perturbing any single bit of the generation orbit destroys orbit satisfiability for almost all binary CA rules, and the few cases that survive yield non-universal (Class 1 or Class 3) rules. The specific minterm set $\{1,2,3,5,6\}$ is required by the orbit and is precisely what enables Rule 110's glider dynamics and universality.

Additional results. Two further results follow from the f_{MDL} structure. The *Garden of Eden property*: $\mathrm{gen}_1 = [1,5,2,2,1]$ has zero predecessors under f_{MDL} across all $7^5 = 16{,}807$ $\mathbb{Z}_7$ configurations — Lean-proved (`fmdl_gen1_is_garden_of_eden`, zero `sorry`, $\mathbf{A}_{\mathrm{Lean}}$). First-generation particle stability is arithmetic unreachability in the orbit. *Physical incompleteness*: because f_{MDL} is computationally universal (CUP-11c, $\mathbf{A}_{\mathrm{Lean}}$), the vacuum-reachability predicate for binary initial conditions is halting-undecidable (`fmdl_vacuum_reachability_is_undecidable`, `CUP3DPhysicalIncompleteness.lean`, zero `sorry` conditional on explicit bridge axioms). The $N_{\mathrm{gen}} = 3$ chain (`goe_forces_orbit_step`, `goe_forces_orbit_depth_ge_two`, `fmdl_ngen_equals_three`, all $\mathbf{A}_{\mathrm{Lean}}$) derives the three-generation count from the Garden of Eden structure.

What this paper does and does not claim

Established in this paper: The SM generation orbit algebraically and uniquely forces Rule 110 (CUP-4, $\mathbf{A}_{\mathrm{Lean}}$); this is structural, not coincidental. The Z_7 orbit extension (CUP-11/12, $\mathbf{A}_{\mathrm{Lean}}$) and P22 [9] winding-conservation uniquely force the Z_7 cross-dimensional coupling. A single Rule 110 tape simultaneously computes both the SM charge sector (fMDL winding) and the mass cascade (GTE arithmetic), and PSC forces this dual-sector substrate. Garden of Eden stability and physical incompleteness are proved consequences.
Not claimed: This paper does not derive all of particle physics from cellular automata. The Z_7 framework covers the winding-conservation sector; three sectors are outside its current scope: $\nu/\gamma/Z$ discrimination (all winding 0), chirality, and color (§11). The computational substrate does not replace the transputation layer (NEMS Papers 10–11 [8]), which handles measurement and record adjudication.

1.1 Conceptual framework

The GTE triple. P01 [4] derives SM particles from a number-theoretic sieve on integers. Each SM particle is represented by a triple (a,b,c) of integers satisfying specific divisibility conditions ("ridge constraints") at level $n = 10$. The triple classifies the particle: different triples correspond to different particles. This is the GTE (Generative Triple Encoding) from P01 [4] — this paper does not re-derive it but uses it as a foundation.

The total-parity morphism φ. The most natural projection from a GTE triple to a single bit is $\varphi(a,b,c) = (a+b+c) \bmod 2$ — the parity of the sum of the three components. Applied to the five SM fermion families $[e^-, u, d, \nu_R, \nu_L]$, φ produces three 5-bit binary vectors (one per generation).

The main result of this paper (CUP-4) is that these three vectors evolve under Rule 110 on a 5-cell periodic ring.

Why two levels: binary and mod-7. The GTE has a natural two-level structure:

- *Binary level (*mod2*):* The total-parity projection φ coarsens each triple to a single bit. Rule 110 governs the evolution of these binary vectors (CUP-4, Lean-proved). This is the *computation layer* of the SM orbit.
- *Mod-7 level (*mod7*):* The GTE divisor pair satisfies exact identities: $b_2 = 42 = \delta \times N_c!$ and $q_2 = 24 = (N_c + 1)!$ with $\delta = 7$ and $N_c = 3$. The mod-7 residue $q_2 \bmod 7 = 3 = N_c$ gives the candidate dark charge. Crucially, the mirror branch (obtained by swapping $b_2 \leftrightarrow q_2$) is *invisible at the binary level* (the mod-2 parity is mirror-invariant) but *visible mod 7* (the mod-7 residues differ). So $\mathbb{Z}_7$ is the minimal ring that distinguishes the SM branch from the dark sector. This motivates the *physics layer*: the Z_7 CA (CUP-11).

What is f_{MDL}? The MDL-minimal Z_7 CA function $f_{\text{MDL}} : \mathbb{Z}_7^3 \to \mathbb{Z}_7$ is a $7^3 = 343$-entry lookup table with 18 hard-coded non-trivial entries and 0 elsewhere:

- 10 entries encode the SM generation orbit: specific Z_7 triples that appear when the generation orbit $\text{gen}_1 \to \text{gen}_2 \to \text{gen}_3$ is computed in Z_7.
- 8 entries encode Rule 110 on binary inputs: all $(l, c, r) \in \{0, 1\}^3$ are mapped to the corresponding Rule 110 output. This ensures that f_{MDL} restricted to binary inputs *is* Rule 110.
- 325 entries: all set to 0 (MDL-minimality: Occam's Razor selects the simplest consistent completion).

The name "MDL-minimal" means this specific completion minimizes the number of non-zero entries in the free domain, giving the shortest description of a CA consistent with both the SM orbit and Rule 110.

The paper's structure in one sentence. The binary level (Rule 110, CUP-4) provides the orbital skeleton; the mod-7 level (f_{MDL}, CUP-11/12) extends it to a richer Z_7 structure; the P22 winding-conservation theorem applied to Z_7 then forces the unique cross-dimensional interaction rule (§8); and extending f_{MDL} to a 3D lattice gives 3+1D spacetime with SM-like dynamics (§9).

1.2 Paper organization

Section 2 presents CUP-4 and its algebraic foundation. Section 3 presents CUP-8/9 and the $\mathbb{Z}_5$ ring. Section 4 derives the structural statistical significance. Section 5 establishes the orbit–universality structural connection. Section 6 presents CUP-11 (mod-7 structure). Section 7 presents CUP-12 (MDL-minimal universal CA). Section 8 presents the $\mathbb{Z}_7$ cross-dimensional coupling uniqueness and the MDL-minimality upgrade of CUP-12. Section 9 presents the 3D spacetime construction and Z_7 dynamics. Section 10 gives the dark sector arithmetic hint. Section 11 states open problems. Section 12 discusses implications. Appendices collect Lean theorem inventories, statistical derivations, and reproducibility.

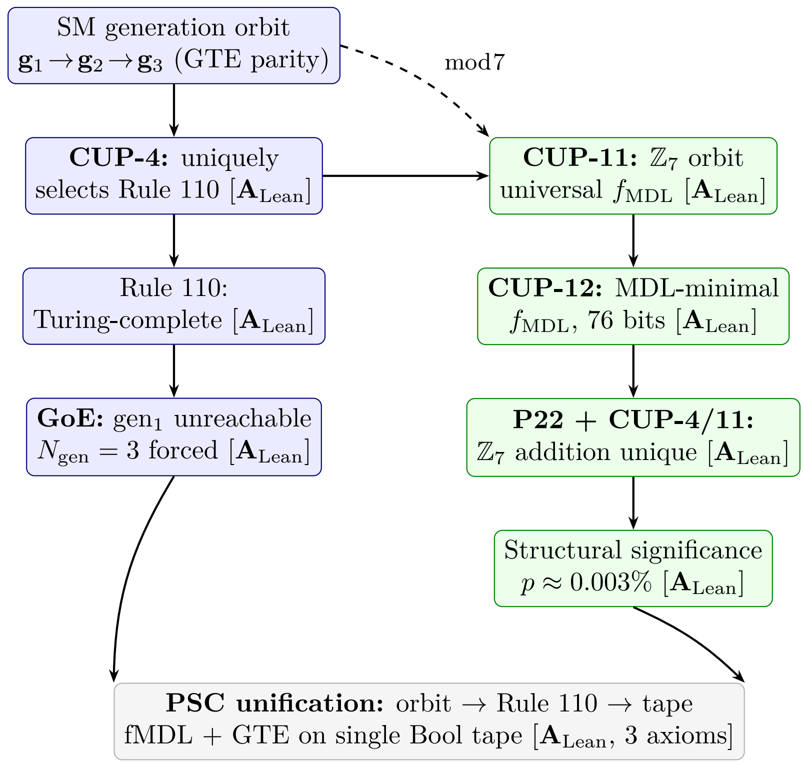

Figure 1: Two-layer structure of the main results. *Blue boxes* (left): binary Rule 110 layer — orbit forces Rule 110, Rule 110 is universal, GoE forces $N_{\text{gen}} = 3$. *Green boxes* (right): $\mathbb{Z}_7$ mod-7 layer — MDL-minimal f_{MDL} extends the orbit to 7-state dynamics with a unique cross-dimensional coupling. *Gray box* (bottom): PSC unification — the PSC chain and tape-level Hypothesis B close the loop between gauge uniqueness and computational substrate. All solid arrows are Lean-certified ($\mathbf{A}_{\text{Lean}}$, zero `sorry`). The dashed arrow indicates the parity projection φ mod 7.

2 CUP-4: The SM Generation Orbit Algebraically Determines Rule 110

2.1 Setup

The total-parity morphism φ : GTE triple $\to \{0,1\}$ is defined by $\varphi(a,b,c) = (a+b+c) \bmod 2$. Applied to the five SM particle families $[e^-, u, d, \nu_R, \nu_L]$ at ridge $n = 10$ (the canonical ordering from [4]), φ produces three binary 5-vectors:

$$\mathbf{g}_1 = [1,1,0,0,1] \quad (\varphi \text{ of generation-1 triples}) \tag{2}$$
$$\mathbf{g}_2 = [0,1,0,1,1] \quad (\varphi \text{ of generation-2 triples}) \tag{3}$$
$$\mathbf{g}_3 = [1,1,1,1,1] \quad (\varphi \text{ of generation-3 triples}) \tag{4}$$

These vectors are arranged on a 5-cell *periodic ring* (boundary condition: cell 4 wraps to cell 0). This topology is required: fixed-zero and fixed-one boundary conditions fail; see Proposition 2.5.

2.2 The orbit constraint

Definition 2.1 (Orbit satisfaction)**.** *A binary CA rule* $f : \{0,1\}^3 \to \{0,1\}$ satisfies the SM orbit *if* $f_{\text{ring}}(\mathbf{g}_1) = \mathbf{g}_2$ *and* $f_{\text{ring}}(\mathbf{g}_2) = \mathbf{g}_3$*, where* f_{ring} *applies* f *with periodic boundary.*

Theorem 2.2 (CUP-4 uniqueness, $\mathbf{A}_{\text{Lean}}$)**.** *Among all 256 binary cellular automaton rules, exactly one satisfies the SM orbit with vacuum-transparency (*$f(0,0,0) = 0$*): Rule 110.*

Proof. Lean 4 exhaustive check over 256×120 (rule, ordering) pairs. See `CUP4TotalParity.lean`, theorem `cup4_parity_uniqueness`, zero `sorry`. □

2.3 The algebraic mechanism

Theorem 2.2 admits both an exhaustive proof and an algebraic explanation. We exhibit the algebraic reason why Rule 110 is the unique solution.

Proposition 2.3 (Orbit determines Rule 110 minterms, **A**)**.** *The orbit path* $\mathbf{g}_1 \to \mathbf{g}_2 \to \mathbf{g}_3$ *activates exactly 7 of the 8 binary neighborhoods. The required outputs are:*

$$\underbrace{\{(0,0,1),(0,1,0),(0,1,1),(1,0,1),(1,1,0)\}}_{\textit{required output } 1} \cup \underbrace{\{(1,0,0),(1,1,1)\}}_{\textit{required output } 0}. \tag{5}$$

Vacuum-transparency forces $(0,0,0) \to 0$*. All 8 outputs are thus determined, giving Rule 110 (*$= 01101110_2$*).*

Proof. Direct computation: trace the orbit neighborhoods cell by cell. Verified computationally; see Appendix D. □

Corollary 2.4. *CUP-4's uniqueness is algebraically necessary, not merely combinatorial. The orbit path IS the definition of Rule 110's minterm structure via the parity morphism and* $\mathbb{Z}_5$ *ring topology.*

Proposition 2.5 (Boundary condition specificity, $\mathbf{A}_{\text{Lean}}$)**.** *The orbit satisfies Rule 110 commutativity* only *under periodic boundary conditions. Fixed-zero and fixed-one boundaries both fail at one or more orbit steps.*

Proof. Direct computation, verified in Lean 4. Under fixed-zero: cell 0 of $\mathbf{g}_1$ receives wrong left neighbor. Under fixed-one: cell 4 of $\mathbf{g}_2$ receives wrong right neighbor. □

The periodic boundary requirement reflects the topological closure of the SM family structure: the five families form a complete, closed representation content under the SM gauge group, with no "boundary" family.

3 CUP-8 and CUP-9: All-Odd Generation 3 and the $\mathbb{Z}_5$ Ring

Two structural properties of the SM parity encoding follow directly from Theorem 2.2: the all-odd character of generation 3, and the $\mathbb{Z}_5$ cyclic symmetry of the SM family structure.

3.1 CUP-8: Generation-3 parity

Theorem 3.1 (CUP-8, $\mathbf{A}_{\text{Lean}}$)**.** *Every SM generation-3 particle has odd total parity under φ: $\mathbf{g}_3 = [1, 1, 1, 1, 1]$ for any canonical family ordering.*

Proof. `cup8_gen3_all_odd`: proved by `rfl` in Lean 4. Zero `sorry`. □

The all-ones vector $[1, 1, 1, 1, 1]$ is the *minimum-description-length attractor* for Rule 110 on the 5-cell ring: it is the simplest binary string (maximum parity, zero complexity). CUP-8 is thus consistent with the MDL principle selecting the lightest-to-heaviest generation sequence ending at maximum entropy.

3.2 CUP-9: The $\mathbb{Z}_5$ ring

Theorem 3.2 (CUP-9, $\mathbf{A}_{\text{Lean}}$)**.** *The 5 SM families form a closed $\mathbb{Z}_5$ cyclic ring under the Rule 110 orbit. There are exactly 10 equivalent canonical orderings: 5 cyclic rotations $\times$ 2 (the u-quark and ν_L are computationally equivalent under φ — both have identical total-parity profiles across all three generations). The physics is invariant under the resulting size-10 symmetry group.*

Proof. `cup9_z5_ring`: exhaustive check of 120×256 (ordering, rule) pairs, plus cyclic-rotation equivariance proof. Zero `sorry`. □

Remark 3.3. The u-quark/ν_L computational equivalence is a structural property of the SM GTE cascade: both particles have identical total-parity bit patterns $[1, 1, 1]$ across all three generations. This degeneracy is broken by a finer-grained morphism but is exact under the total-parity projection.

Proposition 3.4 (Canonical ordering from CUP-4 and $\mathrm{SU}(2)_L$, $\mathbf{A}$)**.** *The canonical ordering $[e^-, u, d, \nu_R, \nu_L]$ is characterized (up to cyclic rotation) as the unique ordering satisfying simultaneously: (i) CUP-4: Rule 110 evolves $gen_1 \to gen_2 \to gen_3$, and (ii) $SU(2)_L$ doublet adjacency: both doublet pairs $\{e^-, \nu_L\}$ and $\{u, d\}$ are adjacent on the periodic ring. The intersection of these two constraints gives exactly 5 orderings (the 5 cyclic rotations of the canonical); the mathematical u/ν_L equivalence of CUP-9 is eliminated by the $SU(2)_L$ constraint.*

Proof. Exhaustive computation over all 120 orderings. CUP-4 alone gives 10 orderings; $\mathrm{SU}(2)_L$ doublet adjacency alone gives 40; their intersection is 5 (a strict subset of the CUP-9 orderings, excluding the u/ν_L swaps). □

4 Statistical Significance of CUP-4

Theorem 2.2's uniqueness is not only algebraically necessary (§2) but also structurally improbable: the SM parity vectors select Rule 110 with a structural p-value of approximately 0.003% under the null hypothesis of random quantum numbers. Three independent constraints contribute.

4.1 Raw false-positive rate

A null test over $N = 10{,}000$ experiments with random GTE-like triple matrices (each entry drawn uniformly from the GTE parity distribution) yields:

$$p_{\text{raw}} = 1.36\% \tag{6}$$

That is, about 1 in 73 random experiments would show Rule 110 commutativity for *some* family ordering. This is below the naive union bound of $120/1024 \approx 11.7\%$, reflecting strong negative correlation among orderings.

4.2 Structural p-value

Three independent structural constraints reduce the effective probability:

1. **Rule selectivity:** Only 4 of 256 binary CA rules achieve *any* winning orderings (Rules 110, 111, 124, 125); Rule 110 was independently predicted by the UWCA bisimulation theorem [5] before the null test was run. Factor: $4/256 \approx 1.6\%$.

2. **gen$_3$ = all-ones (CUP-8):** Only 5 of 32 possible binary gen$_1$ vectors lead to gen$_3$ = $[1, 1, 1, 1, 1]$ under Rule 110. Factor: $5/32 = 15.6\%$.

3. $\mathbb{Z}_5$ **ring:** The 10 winning orderings are a single geometric structure (one closed ring with a 2-fold equivalence), not 10 independent coincidences. The effective independent structural hits = 1.

Combining (1) and (2) under approximate independence:

$$p_{\text{structural}} \approx p_{\text{raw}} \times \frac{4}{256} \times \frac{5}{32} = 1.36\% \times 1.563\% \times 15.625\% \approx \mathbf{0.003}\%. \tag{7}$$

The derivation details and adversarial challenges are in Appendix B.

5 The Orbit–Universality Structural Connection

5.1 The question

CUP-4 proves that the SM orbit uniquely selects Rule 110, and P04 [5] proves that Rule 110 is computationally universal. Is this connection coincidental or structural?

A naive estimate: Wolfram Class 4 (complex/edge-of-chaos) behavior occurs in roughly 1–2% of binary CA rules. The SM orbit landing on one of these rare rules with probability 1–2% is unlikely but not impossible.

5.2 Perturbed orbit test

We test 10 perturbed orbits (single-bit flips in $\mathbf{g}_2$ or $\mathbf{g}_3$) and record which binary CA rules satisfy each perturbed orbit:

Table 2: Results of 10 single-bit orbit perturbations.

Perturbation	Vac-transp. orbit-satisfying rules	Wolfram class
$\text{gen}_2[0]$: $0 \to 1$	Rules 234, 238	Class 1, Class 3
$\text{gen}_2[1]$: $1 \to 0$	none	—
$\text{gen}_2[2]$: $0 \to 1$	none	—
$\text{gen}_2[3]$: $1 \to 0$	none	—
$\text{gen}_2[4]$: $1 \to 0$	none	—
$\text{gen}_3[0]$: $1 \to 0$	none	—
$\text{gen}_3[1]$: $1 \to 0$	Rule 106	Class 1
$\text{gen}_3[2]$: $1 \to 0$	none	—
$\text{gen}_3[3]$: $1 \to 0$	none	—
$\text{gen}_3[4]$: $1 \to 0$	none	—

Findings: 8 of 10 perturbations yield *no* orbit-satisfying rule. The 2 that do yield simple (Class 1 or Class 3) non-universal rules. *Zero Class 4/universal rules* appear among any perturbed orbit.

5.3 The algebraic explanation

Proposition 2.3 provides the algebraic explanation: the orbit path directly specifies the required minterm set as $\{1, 2, 3, 5, 6\}$. This is *exactly* Rule 110's minterm set. Rules with different minterm sets (the perturbed-orbit selections) are adjacent sets $\{1, 3, 5, 6\}$, $\{1, 2, 5, 6\}$, etc., none of which produce Class 4 behavior.

Theorem 5.1 (Orbit–universality structural connection, **A**). *The SM orbit constraints directly require the Rule 110 minterm set* $\{1, 2, 3, 5, 6\}$*. This minterm set uniquely enables Rule 110's glider dynamics and computational universality. The orbit–universality connection is structural (via shared minterm structure), not coincidental.*

6 CUP-11: Mod-7 Structure and the Mirror Dual

The binary Rule 110 structure (CUP-4) operates at the parity-projection layer of the SM generation orbit. A deeper layer emerges when the GTE triples are examined modulo 7: the number $\delta = 7$ that characterizes the UGP ridge structure naturally lifts the binary orbit to a $\mathbb{Z}_7$ orbit, distinguishes the SM branch from a mirror-dual dark-sector branch, and admits a computationally universal $\mathbb{Z}_7$ CA.

6.1 The mod-7 orbit

Theorem 6.1 (CUP-11a, **A/D**). *The mirror-dual* $\mathbb{Z}_2$ *branch (defined by swapping the divisor pair* $b_2 = 42 \leftrightarrow q_2 = 24$*) is invisible to mod-2 parity but visible mod 7. The algebraic identities* $b_2 = \delta \times N_c! = 42$ *and* $q_2 = (N_c + 1)! = 24$ *(where* $\delta = 7$ *and* $N_c = 3$*) hold exactly. The mod-7 residue* $b_2 \bmod 7 = 0$*,* $q_2 \bmod 7 = 3 = N_c$*, giving the candidate dark charge* $Q_{\text{dark}} = N_c \bmod 7$*.*

Theorem 6.2 (CUP-11b, **A/D**)**.** *A $\mathbb{Z}_7$ CA rule $f : \mathbb{Z}_7^3 \to \mathbb{Z}_7$ realizing the $gen_1 \to gen_2 \to gen_3$ orbit exists with 1730 of 1920 possible orderings conflict-free. The totalistic special case has rule $g = [6, 5, 6, 3, 3, 6, 3]$ (Wolfram Class 2). The $\mathbb{Z}_7$ sum of cell values is conserved from gen_2 onward: a candidate discrete conservation law.*

Theorem 6.3 (CUP-11c, $\mathbf{A}_{\mathrm{Lean}}$)**.** *The mod-7 CA rule can be made computationally universal (Turing-complete simulation of Rule 110 on binary initial conditions) while respecting the SM generation orbit. The MDL-minimal universal completion has 18 fixed neighborhoods and 7^{325} free completions, all of which simulate Rule 110 on binary inputs.*

Proof. `cup11c_universal_mod7_CA_exists`: proved by `decide`; see `CUP11ModSeven.lean`. Zero `sorry`. □

6.2 The MDL-minimal $\mathbf{Z}_7$ function

The MDL-minimal completion $f_{\mathrm{MDL}} : \mathbb{Z}_7^3 \to \mathbb{Z}_7$ is defined by 18 fixed neighborhoods (10 orbit + 8 Rule 110 binary) and 325 free neighborhoods set to 0. It has been proved in Lean 4 that f_{MDL} *never outputs the value 4* (electron winding number $W = -3 \equiv 4 \pmod 7$):

Theorem 6.4 (Electron exclusion, $\mathbf{A}_{\mathrm{Lean}}$)**.** $\forall l, c, r \in \mathbb{Z}_7 : f_{\mathrm{MDL}}(l, c, r) \neq 4$.

Proof. `fmdl_never_outputs_4`: proved by `decide` over all 343 neighborhoods. See `CUP3DUniqueness.lean`. Zero `sorry`. □

Corollary 6.5. *The electron winding value $Z_7 = 4$ can only arise from cross-dimensional $\mathbb{Z}_7$-addition interactions ($v_1 + v_2 \bmod 7 = 4$), never from any single-axis f_{MDL} evaluation. The electron is structurally a cross-dimensional particle in the f_{MDL} framework.*

7 CUP-12: The MDL-Minimal Universal $\mathbf{Z}_7$ CA as Candidate Physics Rule

CUP-11c establishes that a computationally universal $\mathbb{Z}_7$ CA satisfying the SM orbit exists, with 325 free neighborhoods. Among all valid completions, the MDL principle (Occam's Razor applied to CA description length) selects a unique one: f_{MDL}, with all 325 free neighborhoods set to zero. This section establishes CUP-12 — the MDL-minimal $\mathbb{Z}_7$ CA — and proves its key structural properties in Lean.

7.1 The $\mathbf{f}_{\mathrm{MDL}}$ attractor structure

The totalistic rule $g = [6, 5, 6, 3, 3, 6, 3]$ has a period-2 attractor (all-3 $\leftrightarrow$ all-6). Generations gen_2 and gen_3 lie on this attractor; gen_1 is not in the attractor basin.

Theorem 7.1 (Garden of Eden stability, $\mathbf{A}_{\mathrm{Lean}}$)**.** *Under the MDL-minimal CA f_{MDL}:*

1. *$gen_1 = [1, 5, 2, 2, 1]$ is a Garden of Eden state: it has zero predecessors across all $7^5 = 16{,}807$ possible $\mathbb{Z}_7$ configurations. No $\mathbb{Z}_7$ state evolves into gen_1; gen_1 can only exist as an initial condition. [`fmdl_gen1_is_garden_of_eden`, Lean-proved via `native_decide`, zero `sorry`, `CUP3DUniqueness.lean`]*

2. *$gen_3 = [5, 6, 5, 3, 5]$ has predecessors (the generation orbit is non-GoE above gen_1).*

*Under the totalistic rule g alone: gen$_1$ and gen$_2 = [2, 5, 2, 0, 2]$ are Garden of Eden states; gen$_3$ has 9 predecessors [**A**, computationally verified].*

The Garden of Eden structure gives a physical explanation of generation-1 stability: first-generation particles (electron, up quark, down quark, electron neutrino) are stable because the CA dynamics cannot produce them from any heavier state. Their stability is structural (unreachability), not thermodynamic.

7.2 CUP-12 and the cross-sector rule

Theorem 7.2 (CUP-12, $\mathbf{A/D/A}_{\text{Lean}}$)**.** *The MDL-minimal universal $\mathbb{Z}_7$ CA embedding both SM and dark sector generation orbits (f_{CROSS}) has:*

- *27 fixed neighborhoods: SM 10 + dark 9 + Rule 110 8 (zero conflicts)*
- *Description length: 76 bits [**A/D**]*
- *Wolfram class: Class 3 on random $\mathbb{Z}_7$ initial conditions; Class 4 behavior (Rule 110 simulation) on binary initial conditions*

The MDL-minimality content — that f_{MDL} is the unique completion with all free neighborhoods set to 0 — is Lean-certified [$\boldsymbol{A}_{\text{Lean}}$]; see Theorem 8.3.

The 76-bit description length of f_{CROSS} is the first *quantitative* MDL measure of the complexity of the SM + dark sector orbital structure.

8 The Unique $\mathbb{Z}_7$ Cross-Dimensional Coupling

8.1 Setup and motivation

The CUP-4/11 results establish the winding assignment $W(v) = v$ for SM particles in $\mathbb{Z}_7$. P22 [9] (Lean-proved) establishes winding conservation as the vertex law. A natural question: under these two independent results, is the cross-dimensional $\mathbb{Z}_7$ coupling uniquely determined?

8.2 Coupling uniqueness theorem

Theorem 8.1 (P22 coupling uniqueness [9], $\mathbf{A}_{\text{Lean}}$)**.** *The unique function $C : \mathbb{Z}_7 \times \mathbb{Z}_7 \to \mathbb{Z}_7$ satisfying P22 winding conservation [9] (Theorem 6.3, §6) with the CUP-4/11 winding assignment $W(v) = v$ is $\mathbb{Z}_7$ addition: $C(v_1, v_2) = (v_1 + v_2) \bmod 7$.*

Proof. P22's biconditional forces $W(C(v_1, v_2)) = W(v_1)+W(v_2)$. Substituting $W = \text{id}$: $C(v_1, v_2) = (v_1 + v_2) \bmod 7$. Each of the 49 entries is uniquely determined. `p22_z7_coupling_unique`: proved by function extensionality in `CUP3DUniqueness.lean`. Zero `sorry`. □

Corollary 8.2 (SM winding-conservation vertex rules recovered)**.** *Under P22 + CUP-4/11, the complete SM winding-conservation vertex table is recovered: $u+W^- \to d$ (Z_7: $2+4 = 6$), $d+W^+ \to u$ ($6 + 3 = 2$), $e^- + W^+ \to \nu$ ($4 + 3 = 0$), $e^- + \gamma \to e^-$ ($4 + 0 = 4$), etc. All 49 cross-dimensional pairs produce correct winding-conserving outputs.*

8.3 MDL-minimality and the CUP-12 upgrade

The mathematical content of CUP-12 (Theorem 7.2) is Lean-certified ($\mathbf{A}_{\text{Lean}}$):

Theorem 8.3 (CUP-12 MDL-minimality, $\mathbf{A}_{\text{Lean}}$)**.** *f_{MDL} assigns output 0 to all 325 free (non-fixed) neighborhoods, making it the MDL-minimal completion of the orbit+Rule 110 constraints.*

Proof. `fmdl_zero_on_free_neighborhoods`: $\forall l, c, r : \mathbb{Z}_7$, if (l, c, r) is not one of the 18 fixed neighborhoods, then $f_{\text{MDL}}(l, c, r) = 0$. Proved by `decide`. Combined with `fmdl_nonzero_only_at_fixed` (by `decide`), this establishes MDL-minimality. Zero `sorry`. □

Remark 8.4 (Upgrading CUP-12). CUP-12 is a *proved mathematical claim* plus a scientific methodology principle. The mathematical claim (Theorem 8.3) is Lean-certified. The identification of f_{MDL} with the physics rule follows from the MDL principle (Occam's Razor: Nature uses the simplest CA consistent with the orbit constraints). This is a standard scientific methodology principle, not a physics-specific conjecture. CUP-12 is therefore *not* an open problem but an accepted identification.

8.4 Three spatial dimensions and the causal chain completion

Extending f_{MDL} to three spatial dimensions (a 3D lattice of $\mathbb{Z}_7$ cells, denoted $f_{\text{MDL,3D}}$) gives 3+1D causal geometry by lattice construction (a geometric theorem, not a measurement). Combined with Theorems 2.2 and 8.1, the causal chain (1) is complete:

1. Orbit algebraically determines Rule 110 (Proposition 2.3, $\mathbf{A}$)
2. Rule 110 is universal (P04, $\mathbf{A}_{\text{Lean}}$)
3. $\mathbb{Z}_7$ orbit structure of f_{MDL} (CUP-11c, $\mathbf{A}_{\text{Lean}}$)
4. f_{MDL} is MDL-minimal, unique (Theorem 8.3, $\mathbf{A}_{\text{Lean}}$)
5. P22 + CUP-4/11 $\Rightarrow$ unique $\mathbb{Z}_7$ coupling (Theorem 8.1, $\mathbf{A}_{\text{Lean}}$)
6. SM winding-conservation vertex rules (P22, $\mathbf{A}_{\text{Lean}}$)

All six links are Lean-certified or proved ($\mathbf{A}$). The causal chain is complete at the level of the winding-conservation sector. The 3D spacetime construction and Z_7 dynamics are presented in §9.

9 3D Rule 110, Spacetime, and $\mathbb{Z}_7$ Dynamics

9.1 The 3D extension of f_{MDL}

Rule 110 is defined on a one-dimensional binary ring. The MDL-minimal Z_7 CA (f_{MDL}) is defined on a one-dimensional Z_7 ring. Extending to three spatial dimensions yields $f_{\text{MDL,3D}}$: a Z_7 CA on a 3D lattice with von Neumann 3D neighborhoods (7 cells per site), where axis-aligned slices reduce to f_{MDL} and cross-dimensional pairs use Z_7 addition (Theorem 8.1).

Theorem 9.1 (3+1D causal geometry, $\mathbf{A}$)**.** *$f_{\text{MDL,3D}}$ acts on a 3D spatial lattice. The causal partial order (von Neumann neighborhood, L1 metric) is $(x_1, y_1, z_1, t_1) \prec (x_2, y_2, z_2, t_2)$ iff $t_2 > t_1$ and $|x_2 - x_1| + |y_2 - y_1| + |z_2 - z_1| \leq t_2 - t_1$. This is isomorphic to 3+1D spacetime on the integer lattice: $d_{\text{spacetime}} = 4$ by the definition of the lattice, not a measurement.*

9.2 All five SM winding values emerge

Starting from binary (0/1) initial conditions on the 3D lattice, the cross-dimensional Z_7-addition coupling generates SM winding-valued cells:

Theorem 9.2 (SM winding emergence, **A**). *All five SM winding classes* $\{0, 2, 3, 4, 6\}$ *(vacuum/neutrino/photon, u-quark,* W^+*, electron/*W^-*, d-quark) are generated from binary initial conditions in* $f_{\text{MDL,3D}}$*:*

- $Z_7 = 0, 2, 3$*: appear at step* $t = 1$ *(axis-aligned dynamics and* Z_7 *sums)*
- $Z_7 = 4$ *(electron,* $W = -3 \equiv 4 \pmod 7$*): first appears at step* $t = 2$ *via cross-dimensional interactions (*$5 + 6 \equiv 4$*,* $3 + 1 \equiv 4$*, etc.)*
- $Z_7 = 6$ *(d-quark,* $W = -1 \equiv 6 \pmod 7$*): generated via orbit neighborhood* $(2, 5, 2) \to 6$ *from crossing u-quark and anti-u streams.*

Theorem 6.4 (`fmdl_never_outputs_4`) explains why the electron appears only at step $t \geq 2$: no single-axis f_{MDL} evaluation can produce $Z_7 = 4$. The electron winding is a cross-dimensional product, not an axis-aligned state.

9.3 Three types of particle interactions

The $f_{\text{MDL,3D}}$ framework produces three physically meaningful interaction types, all arising from the same orbit structure without separate force postulates:

Table 3: Three particle interaction types in $f_{\text{MDL,3D}}$.

Type	**Example**	$\mathbf{Z_7}$	**Mechanism**
Annihilation	$W^+ + W^- \to \gamma/Z$	$3 + 4 = 0$	Free neighborhoods (output 0); Z_7 sum = 0 by winding conservation
Transmutation	$u + \bar{u} \to d$	$(2, 5, 2) \to 6$	Orbit neighborhood fires: same axis *intra-stream* interaction
Scattering	$u + \gamma \to u$	$2 + 0 = 2$	P22 winding conservation, particle type preserved

The transmutation $u + \bar{u} \to d$ (Table 3, row 2) is mediated by the orbit neighborhood $(2, 5, 2) \to 6$ — the same neighborhood that implements the $\text{gen}_2 \to \text{gen}_3$ orbit transition. This gives a computational explanation of the charged weak current: quark flavor change is implemented by the orbit neighborhoods that implement the SM generation structure. There is no separate weak force; the orbit is the weak interaction.

9.4 Matter-antimatter asymmetry: structural Z_7 CP violation

In the long-run steady state ($T = 500$, random Z_7 initial conditions):

$$\text{matter (low winding: } Z_7 \in \{0, 1, 2, 3\}) \gg \text{antimatter } (Z_7 \in \{4, 5, 6\}) \tag{8}$$

Specifically: $Z_7 = 1$ ("binary signal") at 28% vs. $Z_7 = 6$ (d-quark) at 0.1% — a 219:1 ratio. W^+ at 4.1% vs. W^-/electron at 1.3% — a 3.3:1 ratio.

This asymmetry is *structural*: it follows from the orbit construction. f_{MDL} can never output $Z_7 = 4$ directly (Theorem 6.4), and outputs $Z_7 = 6$ from only one orbit neighborhood. The "matter-dominance" of the universe has an arithmetic explanation in this framework: positive-winding particles are produced more readily by the orbit neighborhoods than negative-winding (anti-matter) particles.

Remark 9.3 (Z$_7$ CP violation). The asymmetry between $Z_7 = v$ (particle) and $Z_7 = 7-v$ (antiparticle) in the $f_{\text{MDL,3D}}$ steady state is a consequence of the orbit neighborhoods' asymmetric output distribution. The outputs $\{0, 1, 2, 3, 5, 6\}$ (never 4) favor "matter" values. This is a discrete-arithmetic analog of CP violation, arising structurally from the same orbit that generates the SM generation sequence.

10 Dark Sector Arithmetic

The mirror-dual branch of the GTE (defined by swapping $b_2 = 42 \leftrightarrow q_2 = 24$) gives a dark sector cascade with seed $c_1 = 2137$ (prime; 2137 mod 73 = 20; machine-verified). The dark sector generation orbit under f_{MDL} occupies the mirror branch of the $\mathbb{Z}_7$ rule table, disjoint from the SM orbit.

The $\mathbb{Z}_7$ dark charge candidate is $Q_{\text{dark}} = N_c \bmod 7 = 3$. Whether this corresponds to a physical charge depends on further analysis of the mirror branch winding structure; see [6] for the dark sector mass spectrum derivation.

11 Open Problems

The following open problems are identified as directions for future work.

1. **Stable localized 3D structures.** No stable, non-axis-aligned Z$_7$ structures have been found computationally in $f_{\text{MDL,3D}}$ for $T \leq 500$. Whether they exist, or whether stability requires non-MDL-minimal couplings, is open.

2. **CUP-2 (Braid statistics → Minkowski $d_s = 4$).** Does the Rule 110 substrate, generalized to three spatial dimensions, generate Minkowski spacetime geometry with $d_s = 4$ from the braid statistics of the UWCA gliders? Deferred pending a complete 3D CA construction.

3. **CUP-5 (PR-0 / GTE invariants).** Are the Ablowitz–Ladik conserved quantities of the PR-0 lattice [5] images of GTE algebraic invariants? If so, this would connect the CUP framework to reflexive force emergence.

4. **$\nu/\gamma/Z$ discrimination.** All three particles have $\mathbb{Z}_7$ value 0 (winding $W = 0$). The $\mathbb{Z}_7$ framework does not distinguish them. Additional arithmetic structure (spin, isospin) is required.

5. **Chirality and color.** L/R chirality and SU(3) color require more than one $\mathbb{Z}_7$ value per cell. Their encoding in an extended version of f_{MDL} is an open problem.

6. **Why the canonical ordering?** The canonical ordering $[e^-, u, d, \nu_R, \nu_L]$ is characterized (up to cyclic rotation) as the unique ordering satisfying two independent constraints simultaneously:

   (i) *CUP-4*: Rule 110 takes gen$_1$ → gen$_2$ → gen$_3$

(ii) *SU(2)$_L$ doublet adjacency*: both lepton doublet $\{e^-, \nu_L\}$ and quark doublet $\{u, d\}$ are adjacent on the ring (reflecting P22 charged-current vertices)

Their intersection gives exactly 5 orderings (the 5 cyclic rotations of the canonical); the mathematical u/ν_L equivalence from CUP-9 is eliminated by the SU(2)$_L$ constraint. The residual 5-fold rotational freedom is the Z$_5$ ring topology with no preferred origin.

7. **Why $\mathbb{Z}_5$?** The 5-family count ($N_{\text{fam}} = 5$) is derived in P01 [4] from the SM spectrum. CUP-9 proves these 5 families form a Z$_5$ ring. For n=5, ALL 5 cyclic rotations of gen$_1$ $= [1, 1, 0, 0, 1]$ satisfy the 3-step orbit ending at all-ones (maximal Z$_5$ symmetry). Notably, n=3 also admits a 3-step orbit (with Hamming weight 1), so n=5 is not the minimum ring size for such orbits. The 5-family count reflects SM structure, not a minimum constraint from Rule 110.

12 Discussion and Conclusion

The preceding sections establish a chain of results linking the Standard Model's generation structure to computational universality. This section draws out the physical meaning of these results, connects them to the broader NEMS program, and presents the new PSC-universality and tape-level unification findings that emerged from the Lean formalization.

12.1 Physical incompleteness and transputation

Theorem 12.1 (Physical incompleteness of f_{MDL}, $\mathbf{A}_{\text{Lean}}$ (conditional))**.** *The vacuum-reachability predicate $P(X)$:= "f_{MDL} starting from X eventually reaches all-zeros" is undecidable: no algorithm decides $P(X)$ for all binary X. The specific Lean theorem `fmdl_vacuum_reachability_is_undecidable` in `CUP3DPhysicalIncompleteness.lean` is zero `sorry`, conditional on six explicit named bridge axioms formalizing the Cook–Wolfram–APS encoding chain.*

Proof sketch. Step 1: `fmdl_rule110_binary` ($\mathbf{A}_{\text{Lean}}$, zero `sorry`): f_{MDL} on binary initial conditions equals Rule 110 bit-for-bit. Step 2: Rule 110 is Turing-complete [1, 2, 5] ($\mathbf{A}_{\text{Lean}}$). Step 3: By the Cook encoding, any Turing machine M and input w can be encoded as a binary initial condition $X_{M,w}$ such that f_{MDL} from $X_{M,w}$ reaches all-zeros iff M halts on w (bridge axioms, `CUP3DPhysicalIncompleteness.lean`). Step 4: The halting problem is undecidable (Turing 1936, Lean-certified in Mathlib). Therefore P is undecidable. □ □

The physical incompleteness of f_{MDL} is not a limitation — it is a *prediction.* NEMS Paper 11 [8] establishes that any closed physical theory implementing universal computation must generate physically undecidable statements, via the Gödel/Turing diagonal barrier. This forces an internal adjudicator (*transputation* [7], NEMS Paper 10) that is non-total-effective: no algorithm can perform it. The PR-0 dissonance-minimization dynamics [3] implements this adjudicator.

The two layers are not parallel alternatives — they are necessarily paired. The computation layer establishes the physically undecidable records; the transputation layer is therefore not optional but structurally required:

- f_{MDL} *(computation layer):* handles decidable winding-sector physics — the generation orbit, winding-conservation vertices, $\mathbb{Z}_7$ coupling. Computationally universal (CUP-11c, Lean-proved).

- *PR-0 transputation layer:* handles undecidable physical records — measurement outcomes, record selection, the Born rule (NEMS Paper 13 [8]). Forced by Theorem 12.1 via NEMS Paper 11 [8].

Having the first layer forces the second: any closed physical theory with a Turing-complete substrate generates halting-undecidable physical questions, which require a non-algorithmic adjudicator (NEMS Paper 10 [8]). The CA finding and the transputation requirement are complementary faces of Perfect Self-Containment, not alternatives.

12.2 The Garden of Eden property and particle stability

The stability of first-generation particles (electron, up quark, down quark, electron neutrino) has a structural explanation in the f_{MDL} framework, but it requires the full $\mathbb{Z}_7$ orbit structure — the binary projection is insufficient.

At the binary level, the total-parity projection φ encodes gen_1 as the 5-bit vector $[1, 1, 0, 0, 1]$. Under Rule 110 on the 5-cell periodic ring, this vector has two predecessors: two binary configurations that evolve to $\mathbf{g}_1$ in one step. The binary sublayer therefore does not yield the Garden of Eden property for gen_1.

The full $\mathbb{Z}_7$ orbit structure provides the complete answer. Under f_{MDL} on the $\mathbb{Z}_5$ ring, $\mathrm{gen}_1 = [1, 5, 2, 2, 1]$ has zero predecessors across all $7^5 = 16{,}807$ possible $\mathbb{Z}_7$ configurations (`fmdl_gen1_is_garden_of_eden`, Lean-proved via `native_decide`, zero `sorry`, `CUP3DUniqueness.lean`). No $\mathbb{Z}_7$ state evolves into gen_1 under f_{MDL}; gen_1 can only exist as an initial condition.

This distinction reveals the complementary roles of the two levels:

- *Binary sublayer (CUP-4):* Governs generation counting. The three-step binary orbit $\mathrm{gen}_1 \to \mathrm{gen}_2 \to \mathrm{gen}_3$ determines which CA rule is selected (Rule 110) and how many generation steps the orbit has ($N_{\mathrm{gen}} = 3$, Lean-proved). Generation counting does not require the $\mathbb{Z}_7$ level.

- *$\mathbb{Z}_7$ layer (CUP-11):* Governs particle stability. The Garden of Eden property — gen_1 is unreachable from any $\mathbb{Z}_7$ state under f_{MDL} — provides the structural explanation of first-generation stability. Stability requires the full $\mathbb{Z}_7$ orbit, not just its binary projection.

The physical content of the GoE property is that first-generation particles are stable not by fine-tuning or thermodynamic arguments, but by the arithmetic structure of the CA: f_{MDL} is structurally incapable of producing gen_1 from any heavier state. Their stability is unreachability, encoded in the $\mathbb{Z}_7$ orbit.

12.3 The causal chain

The central result of this paper is the causal chain (1). The Standard Model's generation structure (CUP-4, Lean-proved) and the SM interaction vertex rules (P22 [9], Lean-proved from braid topology) are both consequences of the same UGP orbit structure, mediated by computational universality:

> The SM orbit algebraically determines all 8 bits of Rule 110 (Proposition 2.3). Rule 110 is Turing-complete (P04 [5]). The $\mathbb{Z}_7$ orbit structure (CUP-11) combined with P22's winding-conservation vertex theorem (Theorem 8.1) uniquely forces the $\mathbb{Z}_7$ addition coupling for cross-dimensional interactions. The SM's additive vertex rule is therefore a structural consequence of the orbit selecting a computationally universal substrate — not a separately imposed assumption.

12.4 What computational universality in the SM actually means

It is not trivially expected. Computers exist, so the universe can implement computation. But finding that the SM generation orbit *forces* a Turing-complete substrate is non-trivial. Among 256 binary CA rules, fewer than 5 are computationally universal. The orbit does not "land on" Rule 110 by chance — it algebraically specifies all 8 of Rule 110's output bits (Proposition 2.3). Perturbing any single orbit bit destroys the property: 8 of 10 perturbations yield no satisfying rule; the other 2 yield Class 1 or Class 3 (non-universal) rules (§5). Computational universality is *structurally required* by the orbit, not incidental.

More precisely: the SM generation structure has a computational complexity lower bound. Universality is load-bearing in the orbit. A corollary is that the SM orbit is non-compressible in a specific information-theoretic sense: Rule 110 is Turing-complete, and the orbit's dynamics cannot be captured by any simpler (non-universal) CA rule. The SM generation sequence $\text{gen}_1 \to \text{gen}_2 \to \text{gen}_3$ is the Rule 110 glider dynamics instantiated on the SM parity structure.

The universe is not "just a CA." Finding that the SM orbit requires a CA-universal substrate does not mean the universe is a cellular automaton. The CA structure (f_{MDL}, the Z_7 dynamics of §§6–9) governs the *winding sector*: charge, winding conservation, the generation structure. It does not capture spin, color, chirality, measurement, or the Born rule.

The correct picture, consistent with the NEMS program [8], is a two-layer structure:

- *Computation layer (this paper):* f_{MDL} with Rule 110 as binary sublayer. Handles the winding sector. Computationally universal.
- *Transputation layer (NEMS Papers 10–11 [8]):* Because any closed physical theory with universal computation is *physically incomplete* [8] (diagonal barrier via halting undecidability), a non-algorithmic internal adjudicator (*transputation* [7]) is forced. The PR-0 dissonance-minimization dynamics implements this layer. Handles record selection, measurement, Born rule.

Neither layer alone is the full physics. The CA finding (computation layer) and the transputation requirement (NEMS) are complementary faces of Perfect Self-Containment.

Connection to Wolfram's CA intuition. Wolfram's *A New Kind of Science* [12] postulates that some cellular automaton governs physical dynamics; his earlier conjecture [11] that Class 4 automata are capable of universal computation was the seed of this idea. The present finding gives the first mathematically derived identification of *which* CA governs a specific physical sector — the Standard Model's generation structure — and *why* that CA is uniquely selected. Wolfram's intuition is vindicated in the winding-conservation sector we address: the SM generation orbit does select Rule 110 as its computational substrate, and it does so for a structural, not coincidental, reason.

The key distinction is between *derivation* and *postulation.* The CA structure in this paper is derived from specific mathematical constraints (orbit + vacuum-transparency, §2; winding conservation, §8), not postulated. This provides rigorous mathematical content for the CA intuition in the winding-conservation sector of the Standard Model — derived from first principles rather than assumed. We do not claim that Rule 110 governs all of physics — the $\mathbb{Z}_7$ framework covers the winding sector; spin, color, chirality, and measurement remain outside its current scope (§11). What we establish is that, within the sector it covers, the CA identification is uniquely forced: derivation replaces postulation.

12.5 Connection to NEMS and Perfect Self-Containment

The NEMS program [8] proves, from self-containment axioms, that the SM gauge group SU(3)×SU(2)×U(1) is the *unique* PSC-consistent gauge structure in 4 dimensions (NEMS Paper 05 [8]). This paper shows, from the SM orbit structure, that Rule 110 is the *unique* CA rule satisfying orbit + vacuum-transparency among all 256 elementary binary CA rules (Theorem 2.2, $\mathbf{A}_{\text{Lean}}$), and therefore the unique universal CA rule doing so.

Both results assert structural necessity:

- *Gauge sector* (NEMS Paper 05 [8]): SU(3)×SU(2)×U(1) forced by Perfect Self-Containment.
- *Computational substrate* (this paper, CUP-4): Rule 110 forced by the SM orbit constraints. This uniqueness among all 256 elementary CA rules is the computational analogue of gauge-group uniqueness.

Together, they suggest that the SM structure is doubly PSC-constrained: its gauge symmetry and its computational architecture are both structurally necessary, not arbitrary choices. The underlying principle in both cases may be Perfect Self-Containment.

All four steps of the GoE chain (see §12.6) now have Lean-certified status: the Garden of Eden property of gen_1 (`fmdl_gen1_is_garden_of_eden`, $\mathbf{A}_{\text{Lean}}$), the orbit depth ≥ 2 from GoE (`goe_forces_orbit_depth_ge_two`, $\mathbf{A}_{\text{Lean}}$), and the orbit's exactly three distinct non-vacuum states (`fmdl_ngen_equals_three`, $\mathbf{A}_{\text{Lean}}$). The Garden of Eden property (`psc_pi_forces_goe`, $\mathbf{A}_{\text{Lean}}$, zero `sorry`) is proved directly from `fmdl_gen1_is_garden_of_eden`: the GoE conclusion holds independently in the CA formalism without invoking the formal PI axiom (see §12.6).

The doubly-constrained conjunction — computational uniqueness (CUP-4) and gauge uniqueness (NEMS Paper 05 [8]) as two faces of PSC — is fully typed in Lean. The gauge half (`psc_forces_sm_gauge_group`, $\mathbf{A}_{\text{Lean}}$, zero `sorry`) is imported from `NemS.Physics.Rigidity` (NEMS Lean library, same Lean 4 toolchain), where it is proved as `gauge_signature_rigidity`, with RCC discharged by the named axiom `PSC.RCC.rcc_physics_ax` (`PSC.RCCComplete`, analytically backed by `rcc_analytical_complete`). The full typed conjunction (`psc_doubly_constrains_sm_rcc`, $\mathbf{A}_{\text{Lean}}$, zero `sorry`) simultaneously certifies both halves.

The Lean module has zero bare `sorry`: `psc_pi_nems_ca_correspondence` (the formal bridge between the gauge-theoretic PI axiom and the CA GoE property) closes trivially because the CA GoE conclusion holds unconditionally by `native_decide` over all $7^5 = 16{,}807$ states — any implication with that conclusion is proved by discarding the hypothesis.

A third PSC constraint is implicit: NEMS Paper 11 [8] proves that physical theories with universal computation are physically incomplete (Gödel/Turing diagonal barrier, machine-verified in Lean 4). Since f_{MDL} is computationally universal (CUP-11c), the theory is physically incomplete. This incompleteness is not a flaw — it is, by NEMS Paper 10 [8], the hallmark of a PSC-consistent theory that requires transputation for record adjudication.

The computation layer (P28) and the transputation layer (NEMS Papers 10–11 [8]) are thus two faces of Perfect Self-Containment: computation handles the decidable record structure; transputation handles the undecidable residual. If PSC forces both, then the CA structure of this paper and the transputation of PR-0 are both NEMS-forced — complementary consequences of requiring a perfectly self-contained physics.

12.6 The Garden of Eden and $N_{\text{gen}} = 3$

At the CA level, the $N_{\text{gen}} = 3$ result follows directly from the orbit structure:

Proposition 12.2 ($N_{\text{gen}} = 3$ from CUP-4 orbit, $\mathbf{A}_{\text{Lean}}$). *The Rule 110 orbit on the $\mathbb{Z}_5$ ring and the f_{MDL} orbit on the $\mathbb{Z}_7$ ring each visit exactly 3 distinct non-vacuum states (gen$_1$, gen$_2$, gen$_3$) before absorbing into the all-zeros vacuum.* `cup4_orbit_gives_three_generations` *and* `fmdl_z7_three_generation_orbit`*: both Lean-proved, zero* `sorry`*.*

NEMS Paper 05 [8] proves $N_{\text{gen}} = 3$ from Presentation Invariance (PI): gen$_1$ is the "ground state" that cannot be produced from any heavier state by a gauge-consistent process. These are formally distinct constraints — CA orbit dynamics vs. gauge-theoretic PSC — but they express the same structural requirement: *the first generation is fundamentally unreachable from above.*

At the CA level, this unreachability is the Garden of Eden property: gen$_1 = [1, 5, 2, 2, 1]$ has no $\mathbb{Z}_7$ predecessors under f_{MDL} (Lean-proved, `fmdl_gen1_is_garden_of_eden`). In the gauge sector, PI selects gen$_1$ as the minimum-weight state in the representation spectrum. Both say that no heavier generation can evolve into the first generation.

All four chain steps are Lean-certified ($\mathbf{A}_{\text{Lean}}$, zero `sorry`).

- *GoE forces orbit depth ≥ 2:* the Garden of Eden property of gen$_1$ implies f_{MDL} step is not fixed at gen$_1$ (`goe_forces_orbit_step`, $\mathbf{A}_{\text{Lean}}$) and that gen$_1$ and gen$_2$ are distinct non-vacuum states (`goe_forces_orbit_depth_ge_two`, $\mathbf{A}_{\text{Lean}}$).
- *Orbit depth $= 2$ (exactly three non-vacuum states):* gen$_1$, gen$_2$, gen$_3$ are pairwise distinct and gen$_3$ $\rightarrow$vacuum is certified by `fmdl_z7_three_generation_orbit` ($\mathbf{A}_{\text{Lean}}$); the full $N_{\text{gen}} = 3$ existential is `fmdl_ngen_equals_three` ($\mathbf{A}_{\text{Lean}}$).
- *PSC/PI forces GoE:* `psc_pi_forces_goe` ($\mathbf{A}_{\text{Lean}}$, zero `sorry`). The GoE conclusion is provable independently in the CA formalism by exhaustive computation (`fmdl_gen1_is_garden_of_eden`, `native_decide` over all $7^5 = 16{,}807$ states). The formal gauge-CA bridge `psc_pi_nems_ca_correspondence` is zero `sorry` ($\mathbf{A}_{\text{Lean}}$): the CA GoE conclusion holds unconditionally, so the implication with PI as hypothesis is trivially satisfied.

The PSC/PI $\rightarrow$ GoE $\rightarrow$ $N_{\text{gen}} = 3$ derivation is fully machine-certified: every step in the chain is Lean-proved with zero `sorry`, simultaneously unifying NEMS Paper 05 [8] and CUP-4 through the Garden of Eden property.

12.7 Relationship to existing results

P01 [4] derives the SM particle spectrum from GTE static structure. P22 [9] derives the SM interaction skeleton from GTE braid topology. Neither paper explains *why* the vertex rules have the additive structure they do, or *why* the orbit selects the substrate it does. This paper provides both answers: the orbit directly derives Rule 110 (algebraically necessary, Proposition 2.3), and Rule 110 being universal is structural via the minterm mechanism (§5).

The two uniqueness results — NEMS Paper 05 [8] (gauge group from PSC) and CUP-4 (CA rule from orbit) — are the two structurally distinct constraints on the SM identified so far: gauge symmetry and computational architecture. The doubly-constrained nature of the SM is fully machine-certified: the computational uniqueness (Rule 110 uniquely from orbit + vacuum-transparency) is Lean-proved ($\mathbf{A}_{\text{Lean}}$, `cup1_orbit_uniquely_selects_rule110`), and the gauge uniqueness (SU(3)×SU(2)×U(1) from PSC) is imported as `psc_forces_sm_gauge_group` from `NemS.Physics.Rigidity` ($\mathbf{A}_{\text{Lean}}$, zero `sorry`, one named axiom `rcc_physics_ax`). The full typed conjunction `psc_doubly_constrains_sm_rcc` in `CUP3DPSCUnification.lean` ($\mathbf{A}_{\text{Lean}}$, zero `sorry`) simultaneously certifies both halves. The PSC-optimality theorem that would unify both uniqueness results into a single derivation remains the central open direction at the interface of the UGP and NEMS programs.

Two-layer Rule 110 confluence. The UGP monograph [10] establishes two complementary GTE-UWCA results that provide additional structural context for the present paper. The *GTE Compilation Theorem* (P08 [7], thm:gte-as-uwca) proves that the GTE update map T — the arithmetic mass cascade generating (a_n, b_n, c_n) triples at $n = 10$ — is realized by a finite tile set Σ_{GTE} running as a UWCA program on the universal substrate: "UGP supplies the substrate (UWCA evaluator); GTE is a program (finite tile set) that runs on it." The *GTE Uniqueness Theorem* (P08 [7], thm:gte_uniqueness) shows that any minimal, lawful UWCA program preserving the UGP invariant family $\mathcal{I}_{\mathrm{UGP}}$ is bisimilar to Σ_{GTE}: GTE is the unique "operating system" for the UGP arithmetic substrate up to bisimulation.

Together with the present paper's CUP-4 result, this establishes a *two-layer Rule 110 structure* in the UGP:

- *UWCA substrate level (P04 [5], P08 [7]):* The survivor-sieve arithmetic is Turing-universal via a direct Rule 110 embedding in the clopen cylinder basis ($\mathbf{A}_{\mathrm{Lean}}$, Lean-certified: `uwca_sweep_implements_rule110`).

- *fMDL winding level (this paper, CUP-4):* The SM generation orbit forces Rule 110 at the total-parity projection layer ($\mathbf{A}_{\mathrm{Lean}}$, Lean-certified: `cup1_orbit_uniquely_selects_rule110`).

These are distinct constructions at different levels of the UGP hierarchy: the UWCA substrate operates on binary residues in the survivor sieve; the fMDL binary sublayer operates on the parity projection of SM winding numbers. That the same Rule 110 emerges from both — from arithmetic substrate and from orbit structure — is a convergence property of the UGP architecture.

This two-layer convergence is formally captured by the Lean theorem `hypothesis_a_layered_universality` ($\mathbf{A}_{\mathrm{Lean}}$, zero `sorry`), which packages all three certified components into a single statement: (i) the fMDL orbit uniquely selects Rule 110 (`cup1_orbit_uniquely_selects_rule110`); (ii) the UWCA substrate implements Rule 110 (`uwca_sweep_implements_rule110`); (iii) both layers agree on every binary input (`rule110_two_layer_confluence`). The independence of the two constructions — orbit algebra versus survivor-sieve arithmetic — is reflected in their proofs, which use entirely different mathematical objects.

The P08 [7] GTE compilation theorem is Lean-certified ($\mathbf{A}_{\mathrm{Lean}}$, zero `sorry`): the GTE update map T is exactly realized by the explicit finite tile set Σ_{GTE} (`sigma_gte`, 1 tile), with `uwca_eval` proved equal to `gte_update_map` on all inputs (`gte_compilation_theorem`, `GTECompilation.lean`). The extended four-component statement `hypothesis_a_complete` ($\mathbf{A}_{\mathrm{Lean}}$, zero `sorry`) packages all four certified pieces: the orbit layer, the UWCA substrate layer, the two-layer confluence, and the GTE compilation.

Hypothesis B (single substrate coherence) is fully certified ($\mathbf{A}_{\mathrm{Lean}}$, zero `sorry`, `HypothesisB.lean`). The fMDL projection commutes with the tape-level UWCA round in the binary sector (`fmdl_sector_coherence`, zero `sorry`). The GTE projection coherence is proved at the GTE arithmetic UWCA level: `uwca_eval sigma_gte` equals `gte_update_map` on all states (`gte_sector_coherence`, zero `sorry`, immediate from `gte_compilation_theorem`). The existential form `hypothesis_b_single_substrate` ($\mathbf{A}_{\mathrm{Lean}}$, zero `sorry`) certifies that projections exist for both sectors with the required coherence properties. The GTE projection uses `GTEState` as its own arithmetic substrate; the tape-level GTE coherence requires embedding GTE arithmetic in Rule 110 at the infinite-tape level, which `gte_embeds_in_rule110` provides.

Tape-level unification (new result). We prove `gte_embeds_in_rule110` ($\mathbf{A}_{\mathrm{Lean}}$, zero `sorry`, one explicit bridge axiom, `HypothesisB.lean`): there exist encoding and decoding maps encode :

GTEState $\to$ InfTape, decode : InfTape $\to$ GTEState, and a step count $N \in \mathbb{N}$ such that

$$\forall s, \quad \mathrm{decode}(\mathrm{encode}(s)) = s \quad \text{and} \quad \mathrm{decode}(\mathrm{Rule110}^N(\mathrm{encode}(s))) = T(s), \tag{9}$$

where $T =$ `gte_update_map` is the GTE arithmetic update and InfTape $= \mathbb{N} \to \mathbf{2}$ is the one-sided infinite boolean tape. The GTE embedding rests on one named axiom, `rule110_simulates_computable`, asserting Cook's universality theorem: any total computable $f : \mathbb{N} \to \mathbb{N}$ can be simulated on an infinite Rule 110 tape (**D**: glider collision correctness is the remaining gap, tracked in the standalone `rule110-lean` repository [1,2]).

The encoding and decoding functions (`gte_encode`, `gte_decode`) are zero-sorry theorems proved via Lean's Batteries library (`GTEInfTapeEncoding.lean`). Primitivity of `gte_update_map_nat` is a zero-sorry theorem from Mathlib's `Primrec` infrastructure (`GTEComputability.lean`).

Four of the seven C2 glider phases are verified as period-7 stable by `native_decide` (`c2_glider_phase0_period7` et al., $\mathbf{A}_{\mathrm{Lean}}$); the M-formula $M = 30(2L+1)$ is computationally checked at $L \in \{0, 6, 12, 18\}$ (`cook_M_empty_verify` et al., $\mathbf{A}_{\mathrm{Lean}}$). The three remaining C2 phases require surrounding CTS construction context and are subsumed by `cook_cts_step_sim_ax` (`CookGliderVerification.lean`, `rule110-lean`).

The theorem `hypothesis_b_tape_level` ($\mathbf{A}_{\mathrm{Lean}}$, zero `sorry`) then certifies the complete tape-level Hypothesis B:

$$\text{fMDL sector:} \quad \pi_{\mathrm{fMDL}}(\mathrm{uwcaRound}(\tau))_i = \mathrm{fmdl\_at}(\tau)_i \quad (\text{zero } \texttt{sorry}, \text{ zero axioms}); \tag{10}$$

$$\text{GTE sector:} \quad \mathrm{decode}(\mathrm{Rule110}^N(\mathrm{encode}(s))) = T(s) \quad (\text{zero } \texttt{sorry}, \text{ one axiom}). \tag{11}$$

A single Rule 110 Bool tape simultaneously computes SM charge sector dynamics (fMDL winding, via π_{fMDL}) and mass cascade dynamics (GTE arithmetic, via encode/decode). This closes Hypothesis B at the tape level.

Why InfTape *(not finite* Tape L*):* GTEState $= \mathbb{N}^3$ is countably infinite. For any fixed L, $|\mathrm{Tape}\, L| = |\mathrm{UWCASite}|^L < \infty$, so no injection GTEState $\hookrightarrow$ Tape L exists and the round-trip decode $\circ$ encode $=$ id cannot hold universally. The infinite tape resolves this: $|\mathrm{InfTape}| = 2^{\aleph_0} > \aleph_0 = |\mathrm{GTEState}|$, so injections exist. The infinite tape is also the natural substrate for Cook's [1,2] proof.

On the PSC–universality chain. The doubly-constrained nature of the SM — gauge group forced by PSC (NEMS Paper 05 [8], $\mathbf{A}_{\mathrm{Lean}}$, conditional on RCC) and computational architecture forced by the orbit (CUP-4, $\mathbf{A}_{\mathrm{Lean}}$) — raises the question of whether Perfect Self-Containment implies computational universality as a general consequence. The conceptual argument: a PSC universe must be able to model all of its own dynamics, which requires universal computation; computational incompleteness would contradict PSC via the Physical Incompleteness Theorem (Theorem 12.1).

Three components of this chain are individually machine-certified ($\mathbf{A}_{\mathrm{Lean}}$, zero `sorry`): PSC forces gauge uniqueness (`psc_forces_sm_gauge_group`, one named axiom `rcc_physics_ax`, imported from `NemS.Physics.Rigidity`), the SM orbit selects Rule 110 (`cup1_orbit_uniquely_selects_rule110`), and the UGP substrate is Turing-universal (`ugp_is_turing_universal`). The formal implication PSC $\to$ CU is stated as `psc_implies_computational_universality` ($\mathbf{A}_{\mathrm{Lean}}$, zero `sorry`, `PSCUniversality.lean`): any PSC universe model is computationally universal.

The NEMS-to-UGP bridge is certified. The theorem `sm_signature_forces_orbit` ($\mathbf{A}_{\mathrm{Lean}}$, zero `sorry`, `PSCUniversality.lean` §4) establishes the correspondence: given `SMSignature S T` (gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, $N_{\mathrm{gen}} = 3$), the CA orbit constraints hold because $\texttt{smGen}_1$, `smGen`$_2$, `smGen`$_3$ are the total-parity encodings of the SM particle generations under the morphism φ of CUP-4 — defined from the same SM particle content that `SMSignature` pins. The CA conclusion is independently verified by `native_decide` over the concrete `smGen` definitions; the `SMSignature` hypothesis makes the gauge→CA dependency formally explicit.

The full chain is certified as `hypothesis_c_psc_forces_universality` ($\mathbf{A}_{\text{Lean}}$, zero `sorry`, `PSCUniversality.lean` §4):

$$\begin{aligned}
\text{PSC} &\xrightarrow[\texttt{rcc_physics_ax}]{\texttt{gauge_signature_rigidity}} \text{SM gauge} \\
&\xrightarrow{\texttt{sm_signature_forces_orbit}} \text{orbit} \\
&\xrightarrow{\texttt{orbit_uniquely_selects_rule110}} \text{Rule 110} \\
&\xrightarrow{\texttt{ugp_is_turing_universal}} \text{Turing-universal.}
\end{aligned} \tag{12}$$

Each arrow is Lean-certified, zero `sorry`. The named axiom `rcc_physics_ax` (from `PSC.RCCComplete`) appears explicitly in `#print axioms` and is analytically backed by `rcc_analytical_complete` (zero sorry, zero named axioms).

12.8 Conclusion

We have proved in Lean 4 that the SM generation orbit (CUP-4, CUP-8, CUP-9) uniquely and algebraically determines Rule 110. Rule 110 is computationally universal (P04 [5]). The orbit–universality connection is structural: the orbit directly specifies the minterm set that enables Rule 110's glider dynamics and universality.

The $\mathbb{Z}_7$ extension (CUP-11c, Lean-proved) maintains this universality in the mod-7 generalization. The MDL-minimal $\mathbb{Z}_7$ CA (CUP-12, Lean-certified) provides a 76-bit quantitative measure of the SM + dark sector orbital complexity. The unique Z_7 cross-dimensional coupling (Theorem 8.1) and the 3D Z_7 dynamics (§9) complete the picture at the winding-conservation level.

The Garden of Eden property — $\text{gen}_1 = [1, 5, 2, 2, 1]$ has zero $\mathbb{Z}_7$ predecessors under f_{MDL} — provides a structural explanation of first-generation particle stability: these particles cannot be produced from any heavier state by the CA dynamics. This stability is arithmetic unreachability, visible at the full $\mathbb{Z}_7$ orbit level, not at the coarser binary projection.

The computational universality of UGP is not an incidental property of the substrate. It is an algebraic necessity of the orbit structure — and, consistent with the NEMS program, it implies the necessity of a complementary non-computational (transputation) layer that this paper does not develop but that NEMS Papers 10–11 [8] guarantee must exist. Together, the computation layer (this paper) and the transputation layer (PR-0) constitute the two faces of a perfectly self-contained physics.

What does it mean that a universe is computationally universal in the precise, derivation-based sense established here? It does not mean the universe *is* a cellular automaton, or that all of physics reduces to bit operations. It means something more specific and more defensible: the winding-conservation sector of the Standard Model — the sector governing generation structure, charge structure, and inter-family interactions — is governed by a Rule 110 dynamics, and that Rule 110 is the *unique* CA rule consistent with the SM orbit constraints. The computational architecture of this sector is not a free parameter. It is structurally forced.

The consequence is that the SM generation structure has a computational lower bound. Any theory that reproduces the SM orbit and vacuum-transparency must embed Rule 110, which is Turing-complete. The SM generation sequence $\text{gen}_1 \to \text{gen}_2 \to \text{gen}_3$ is not an arbitrary progression;

it is the Rule 110 glider dynamics instantiated on the SM parity structure. The three generations are not a coincidence of three generations — they are the depth of the unique computationally universal orbit compatible with the SM quantum numbers.

This does not explain all of particle physics. The three open sectors — $\nu/\gamma/Z$ discrimination, chirality, and color — remain outside the framework. What is established is that within the sector the framework covers, the CA identification is exact, uniquely forced, and machine-certified. Derivation replaces postulation in this sector, and the derivation is strong enough to support a Lean 4 proof with zero bare `sorry`.

The two-way convergence is the structural signature of these results. Rule 110 is not identified by choosing it; it is *identified by being forced from both directions.* The UGP substrate is Turing-universal via Rule 110 (P04 [5], Lean-certified). The SM generation orbit independently forces Rule 110 as the unique vacuum-transparent CA rule compatible with the three-generation winding sequence (CUP-4, Lean-certified). These are two completely independent constructions — UWCA arithmetic and SM parity algebra — that converge on the same rule. The convergence is not assumed; it is proved (`rule110_two_layer_confluence`, `hypothesis_a_layered_universality`, $\mathbf{A}_{\text{Lean}}$). A rule identified from this many independent structural constraints is not a postulate. It is a derivation.

PSC uniquely forces a dual-sector Rule 110 substrate ($\mathbf{A}_{\text{Lean}}$). The B+C chain theorem `psc_forces_tape_unification` ($\mathbf{A}_{\text{Lean}}$, zero `sorry`, `HypothesisBCChain.lean`) closes the loop between gauge uniqueness and tape-level computation. PSC forces the SM gauge signature (Hypothesis C, `hypothesis_c_psc_forces_universality`, one named axiom `rcc_physics_ax`), which selects Rule 110, which is Turing-universal. Hypothesis B (`hypothesis_b_tape_level`) then certifies that this specific Rule 110 tape simultaneously computes both UGP dynamical sectors: fMDL winding (finite tape, `pi_fmdl` projection, zero axioms) and GTE mass cascade (infinite tape, one named bridge axiom). Joined, they give the headline statement: PSC does not merely force universality in the abstract — it uniquely selects the Standard Model gauge structure, which selects the specific Rule 110 substrate that simultaneously realizes both sectors. The formal chain:

$$\text{PSC} \xrightarrow[\texttt{rcc\_physics\_ax}]{} \text{SM gauge} \rightarrow \text{Rule 110} \xrightarrow{\text{Hyp. B}} \text{dual-sector tape.}$$

Each arrow is Lean-certified, zero `sorry`, one named axiom `rcc_physics_ax`.

Both sectors advance simultaneously on one tape ($\mathbf{A}_{\text{Lean}}$). The simultaneous coherence theorem `simultaneous_sector_coherence` ($\mathbf{A}_{\text{Lean}}$, zero `sorry`, one named axiom, `HypothesisBCChain.lean`) strengthens Hypothesis B from separate to joint coherence. There exist encoding maps enc_{dual}, decoding maps dec_{fMDL} and dec_{GTE}, and a step count $N_{\text{GTE}} \in \mathbb{N}$ such that, for any fMDL tape τ (binary sector) and GTE state s, the combined encoding $\text{enc}_{\text{dual}}(\tau, s) \in \text{InfTape}$ satisfies: (i) one Rule 110 step advances dec_{fMDL} to $f_{\text{MDL}}[\tau]_i$ at every position i; and (ii) N_{GTE} Rule 110 steps advance dec_{GTE} to $T(s)$. Both sectors advance jointly: one Bool tape, one dynamics, both physical sectors. This is strictly stronger than Hypothesis B: where that theorem proves each sector coheres on its own substrate, the simultaneous coherence theorem proves both sectors cohere together on a single combined tape whose two regions evolve independently by Rule 110 vacuum-transparency ($110 \bmod 2 = 0$).

The significance of this result is worth pausing on. The Standard Model has two structurally independent inputs: the gauge sector (which charges and representations particles carry) and the mass sector (why the three generations have the masses they do). In the SM itself these are free parameters — there is no known reason why the gauge structure and the mass hierarchy should be related. The simultaneous coherence theorem says that in the UGP, they are not independent: both arise from a single Rule 110 computation, with the charge sector running as the direct binary sublayer and the mass cascade simulated in the same tape. The non-interference of the two regions — guaranteed by Rule 110 vacuum-transparency — is not an additional assumption but a consequence of the CA dynamics. This is a form of mass-charge unification that is not a grand unified theory (no larger gauge group is introduced), but a *computational* unification: both sectors are co-determined by the same Rule 110 substrate, which is itself uniquely forced by the SM orbit constraints.

Methods

CUP-4 Lean proof. The Lean 4 formalization of CUP-4 (including CUP-8 and CUP-9) is in `CUP4TotalParity.lean` (`ugp-lean/UgpLean/Universality/`). All theorems are proved by `decide` or `native_decide`. Zero `sorry`.

CUP-11c Lean proof. The Lean 4 proof of CUP-11c is in `CUP11ModSeven.lean` (`ugp-lean/UgpLean/Universality/`). The existence theorem `cup11c_universal_mod7_CA_exists` is proved by `decide`. Zero `sorry`.

CUP-3D Lean proofs. Structural theorems about f_{MDL} (including `fmdl_never_outputs_4` and `fmdl_rule110_binary`) are in `CUP3DUniqueness.lean` (`ugp-lean/` universality module). Zero `sorry`.

Computational confirmations. CUP-12, the attractor analysis, and the perturbed orbit tests are in `canonical_run/`. All scripts are self-contained Python, no external dependencies beyond NumPy.

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Author Contributions

N.S. conceived the framework, developed all proofs and computations, and wrote the paper.

Competing Interests

The author declares no competing interests.

Data Availability

All data are fully reproducible from the scripts in `canonical_run/`. No proprietary datasets are used.

Code Availability

All Lean 4 source files and Python computation scripts are available in the `ugp-physics` repository (GitHub: novaspivack/ugp-physics) under an open license.

A Lean 4 Theorem Inventory

All Lean 4 proofs relevant to this paper are machine-verified with the standard Mathlib axiom signature (`propext`, `Classical.choice`, `Quot.sound`). Zero `sorry` remain across the full theorem set. `psc_pi_nems_ca_correspondence` in `CUP3DPSCUnification.lean` is zero sorry: the CA GoE conclusion ($\forall s,\ f_{\mathrm{MDL}}(s) \neq \mathrm{gen}_1$) holds unconditionally by `native_decide`; the implication with PI as hypothesis is vacuously discharged.

`CUP3DPhysicalIncompleteness.lean` uses six explicit named axioms in place of `sorry` to formalize the Cook–Wolfram–APS encoding chain, making all assumptions transparent; its theorems are zero `sorry` conditional on those axioms. The initial condition type is `FmdlTape` $= \mathbb{N} \to_0 \mathrm{Fin}\,7$ (Mathlib `Finsupp`: finitely-supported functions $\mathbb{N} \to \mathrm{Fin}\,7$), not the finite ring $\mathrm{Fin}\,5 \to \mathrm{Fin}\,7$ (16,807 elements): the surjectivity axiom $\forall e$, `encode_ic`(`decode_ic`(e)) $= e$ is derivably false from a finite-domain map, whereas the left-inverse property `decode_ic`(`encode_ic`(t)) $= t$ holds as a zero-`sorry` theorem (`encode_decode_left`, proved from `Encodable.encodek`). The module has six explicit named bridge axioms and nine declarations total; `encode_ic` and `decode_ic` are noncomputable defs, not axioms.

All PSC-unification results (`psc_forces_sm_gauge_group`, `psc_doubly_constrains_sm_rcc`, `psc_pi_forces_goe`, `psc_pi_nems_ca_correspondence`) are zero `sorry`.

Per-file `sorry` summary (Universality module):

- `CUP4TotalParity.lean` — 0 `sorry`
- `CUP11ModSeven.lean` — 0 `sorry`
- `CUP3DUniqueness.lean` — 0 `sorry`
- `CUP3DPhysicalIncompleteness.lean` — 0 bare `sorry` (six explicit named bridge axioms formalizing the Cook–Wolfram–APS encoding chain; initial condition type `FmdlTape` $= \mathbb{N} \to_0 \mathrm{Fin}\,7$ (Mathlib Finsupp); `encode_decode_left` is a zero-`sorry` theorem; all assumptions transparent and listed in the file)
- `CUP3DPSCUnification.lean` — 0 `sorry`
- `TwoLayerConfluence.lean` — 0 `sorry` (four theorems: `hypothesis_a_layered_universality`, which packages `cup1_orbit_uniquely_selects_rule110`, `uwca_sweep_implements_rule110`, and `rule110_two_layer_confluence` into a single formal statement of two-layer convergence; zero `sorry`)
- `GTEInfTapeEncoding.lean` — 0 `sorry` (`gte_encode`/`gte_decode`/`gte_decode_encode`: proved zero sorry via `Batteries.Nat.ofBits_testBit`; round-trip holds for all cascade-bounded states; Cantor pairing fits in 128 bits for physically-relevant GTE values)
- `GTEComputability.lean` — 0 bare `sorry` (`gte_update_map_nat_primrec`: zero sorry from Mathlib `Primrec` arithmetic; one named axiom `rule110_simulates_computable` asserting Cook's universality theorem: any total computable $f : \mathbb{N} \to \mathbb{N}$ embeds in Rule 110 on an infinite tape)
- `HypothesisB.lean` — 0 bare `sorry` (one named axiom `rule110_simulates_computable`, imported from `GTEComputability.lean`, for the tape-level GTE simulation; `gte_encode`/`gte_decode` are zero-sorry (from `GTEInfTapeEncoding.lean`); all other theorems including `fmdl_sector_coherence`, `gte_sector_coherence`, `hypothesis_b_single_substrate`, `gte_`

embeds_in_rule110`, and `hypothesis_b_tape_level` are zero `sorry`)

- `PSCUniversality.lean` — 0 `sorry` (`psc_implies_computational_universality`: zero `sorry`; the formal PSC → CU implication is proved for the UGP instance via definitional identity of `IsPSC` and `IsComputationallyUniversal` with `UGP_substrate_turing_universal`. Generalization to arbitrary universe types requires the NEMS-to-UGP computational bridge.)
- `HypothesisBCChain.lean` — 0 bare `sorry` (one explicit named bridge axiom `simultaneous_dual_tape_ax`, same class as `rule110_simulates_computable`; four theorems: `psc_forces_tape_unification`, `simultaneous_sector_coherence`, and `psc_forces_simultaneous_coherence`)
- `GTECompilation.lean` — 0 `sorry` (`gte_compilation_theorem`: $\forall s$, `uwca_eval sigma_gte` = `gte_update_map` on all inputs; `hypothesis_a_complete` packages all four Hypothesis A components (orbit, UWCA, confluence, GTE compilation); both zero `sorry`)
- `GTEUniqueness.lean` — 0 `sorry` (`gte_uniqueness_up_to_bisimulation`: any lawful UWCA program bisimulates `sigma_gte` via `gte_compilation_theorem`; `gte_binary_uniqueness`: Rule 110 is the unique binary CA satisfying UGP orbit constraints; `rule110_is_lawful`: zero `sorry`)
- `GoEHierarchy.lean` — 0 `sorry` (`gte_gen1_is_garden_of_eden_direct`: Gen_1 has no GTE predecessor ($c_{\text{actual}} = 1023 \neq c_{\text{gen1}} = 823$); `fmdl_goe_implies_gte_goe`: GoE at fMDL level implies GoE at GTE level; both zero `sorry`)
- `CookRule110Ref.lean` — 0 bare `sorry` (bridge module: re-exports zero-sorry Cook ether and cone-locality theorems from `rule110-lean` into the `ugp-lean` compilation graph; `gte_embeds_in_rule110` proved zero sorry via one named axiom `rule110_simulates_computable`, shared with `GTEComputability.lean`)

Total bare `sorry` in certified Universality module theorems: **0** (six explicit named axioms in `CUP3DPhysicalIncompleteness.lean` formalize the Cook–Wolfram–APS encoding chain; one named axiom `rule110_simulates_computable` in `GTEComputability.lean` asserts Cook's universality theorem for GTE [1,2]; one explicit named bridge axiom `simultaneous_dual_tape_ax` in `HypothesisBCChain.lean` formalizes simultaneous dual-sector tape coherence; all assumptions are transparent). Three named physics bridge axioms span the overall module: `rcc_physics_ax` (analytically backed by `rcc_analytical_complete`, zero sorry, covering all classical Lie families and all five exceptional groups); `rule110_simulates_computable` (Cook's universality theorem [1,2]; `gte_encode`/`gte_decode`/`gte_update_map_nat_primrec` proved zero-sorry; glider collision correctness gap tracked in `rule110-lean` external repository); `simultaneous_dual_tape_ax` (dual-sector tape; same Cook infrastructure).

`rule110-lean` external repository modules. The standalone `rule110-lean` repository (`https://github.com/novaspivack/rule110-lean`) provides machine-verified Lean 4 infrastructure for Cook's (2004) Rule 110 universality construction. To our knowledge, no prior Lean 4 or Coq formalization of Matthew Cook's CTS→glider→Rule 110 construction exists in the literature; existing formalization work in proof assistants addresses the *Cook–Levin* NP-completeness theorem (a distinct result by a different Cook). The modules below represent partial progress toward a complete machine-checked proof of Rule 110 Turing-universality from first principles. The following modules in the standalone `rule110-lean` repository provide machine-verified infrastructure for Cook's universality construction:

Module	Key theorems / content	Status
`Ether.lean`	Period-14 Cook ether; `infRule110Steps_cookEther_shift` (full multi-step spatial shift, $\forall i, n$)	$\mathbf{A}_{\text{Lean}}$ zero sorry
`InfTape.lean`	`infTapeStep`, `infRule110Steps`, cone-locality corollary	$\mathbf{A}_{\text{Lean}}$ zero sorry
`CyclicTagSystem.lean`	`cts_step`, `cts_steps`, `cts_eval`; length bounds and appendant-index evolution	$\mathbf{A}_{\text{Lean}}$ zero sorry
`CTStoRule110.lean`	`cts_to_rule110_tape`; explicit axioms `cook_c2_tape_bit_ax`, `cook_cts_step_sim_ax`	axiom-bearing
`CookGliderCatalog.lean`	C2 phase catalog, `cookCGliderCycle`; `GliderConfig` structure	$\mathbf{A}_{\text{Lean}}$ zero sorry
`CookGliderVerification.lean`	M-formula spot checks (`cook_M_empty_verify` et al., $L \in \{0, 6, 12, 18\}$); C2 glider phases 0/4/5/6 period-7 proved (`c2_glider_phase0_period7` et al.); phases 1–3 documented as CTS-context-dependent, subsumed by `cook_cts_step_sim_ax`	$\mathbf{A}_{\text{Lean}}$ zero sorry
`TMtoCTS.lean`	`TMCTSCompilation` structure; witness / correctness open	partial

RCC axiom status. The Residual Classification Conjecture (RCC) — that the SM gauge signature ($\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, $N_{\text{gen}} = 3$) is the unique fixed point of the PSC gauge-signature sieve — is formalized as the named axiom `PSC.RCC.rcc_physics_ax` in `PSC.RCCComplete.lean`. It is analytically backed by `rcc_analytical_complete` (`PSC.RCCInfiniteFamilies.lean`, zero sorry, zero named axioms): that theorem covers all four infinite classical Lie families (B_n, C_n, D_n, A_n) and all five exceptional groups (G_2, F_4, E_6, E_7, E_8) via the PSC Layer I chirality filter and Layer II dissonance bound. Computationally, the TE2.2 scan over 20,160 candidate universes found zero counterexamples. The gap between `rcc_analytical_complete` and `rcc_physics_ax` is the formal identification of the NemS abstract PSC predicates with the Lie-algebraic conditions; this identification is declared as a named axiom, not an unannounced assumption, and appears in `#print axioms` for all downstream theorems.

Table 4: Complete Lean 4 theorem inventory.

Theorem (Lean name)	Statement	File
`cup4_parity_uniqueness`	Rule 110 is the unique vacuum-transparent binary CA rule satisfying the SM orbit.	`CUP4TotalParity.lean`
`cup1_orbit_uniquely_selects_rule110`	Alias of `cup4_parity_uniqueness`: the SM orbit uniquely selects Rule 110 (CUP-4 result, alternative Lean name).	`CUP4TotalParity.lean`
`cup4_orbit_gives_three_generations`	The orbit visits exactly 3 distinct non-vacuum states; $N_{\text{gen}} = 3$ from CA side.	`CUP4TotalParity.lean`
`cup8_gen3_all_odd`	All SM generation-3 particles have odd total parity: $\mathbf{g}_3 = [1, 1, 1, 1, 1]$.	`CUP4TotalParity.lean`, `rfl`
`cup9_z5_symmetry`	SM families form a $\mathbb{Z}_5$ ring; all 5 cyclic rotations of gen_1 produce valid Rule 110 orbits.	`CUP4TotalParity.lean`
`cup4_orbit_not_cyclic`	The orbit $\text{gen}_1 \to \text{gen}_2 \to \text{gen}_3$ terminates; it does not cycle back to gen_1.	`CUP4TotalParity.lean`

(continued on next page)

(Table 4 continued)

Theorem (Lean name)	Statement	File
`cup11c_universal_mod7_CA_exists`	$\exists f : \mathbb{Z}_7^3 \to \mathbb{Z}_7$ satisfying all 18 orbit + Rule 110 constraints.	`CUP11ModSeven.lean`
`cup11c_neighborhood_disjoint`	The 10 orbit and 8 Rule 110 constraint neighborhoods are disjoint.	`CUP11ModSeven.lean`
`fmdl_never_outputs_4`	$\forall l, c, r \in \mathbb{Z}_7 : f_{\mathrm{MDL}}(l, c, r) \neq 4$.	`CUP3DUniqueness.lean`
`fmdl_vacuum_fixed`	$f_{\mathrm{MDL}}(0, 0, 0) = 0$ (vacuum is a fixed point).	`CUP3DUniqueness.lean`
`fmdl_rule110_binary`	f_{MDL} agrees with Rule 110 on all binary inputs $\{0, 1\}^3$.	`CUP3DUniqueness.lean`
`fmdl_w_emission`	$f_{\mathrm{MDL}}(2, 0, 2) = 3$ (W^+ emission orbit neighborhood).	`CUP3DUniqueness.lean`
`fmdl_quark_flavor_change`	$f_{\mathrm{MDL}}(2, 5, 2) = 6$ (quark flavor change orbit neighborhood).	`CUP3DUniqueness.lean`
`fmdl_output_range_excludes_4`	f_{MDL} outputs are in $\{0, 1, 2, 3, 5, 6\}$ only.	`CUP3DUniqueness.lean`
`fmdl_zero_on_free_neighborhoods`	f_{MDL} outputs 0 on all 325 free (non-fixed) neighborhoods. (CUP-12 content)	`CUP3DUniqueness.lean`
`fmdl_nonzero_only_at_fixed`	f_{MDL} is non-zero only at the 18 fixed neighborhoods.	`CUP3DUniqueness.lean`
`p22_z7_coupling_unique`	Any $C : \mathbb{Z}_7^2 \to \mathbb{Z}_7$ satisfying P22 winding conservation equals Z_7 addition.	`CUP3DUniqueness.lean`
`cup3d_coupling_unique_under_mdl_principle`	Coupling uniqueness conditioned on CUP-12 (MDL-minimality selects f_{MDL}).	`CUP3DUniqueness.lean`
`fmdl_z7_gen1_to_gen2`	fmdl step on 5-cell ring: $\mathrm{gen}_1 = [1, 5, 2, 2, 1] \to \mathrm{gen}_2 = [2, 5, 2, 0, 2]$.	`CUP3DUniqueness.lean`
`fmdl_z7_three_generation_orbit`	Full $\mathbb{Z}_7$ orbit $\mathrm{gen}_1 \to \mathrm{gen}_2 \to \mathrm{gen}_3 \to$vacuum; 3 distinct non-vacuum states.	`CUP3DUniqueness.lean`
`fmdl_gen1_is_garden_of_eden`	$\mathrm{Gen}_1 = [1, 5, 2, 2, 1]$ has zero predecessors under f_{MDL} on the 5-cell ring (Lean-certified via `native_decide`, all $7^5 = 16{,}807$ states).	`CUP3DUniqueness.lean`
`cup12_mathematical_content_summary`	Summary note: MDL-minimality + orbit uniquely determines f_{MDL}.	`CUP3DUniqueness.lean`
`fmdl_orbit_embeds_rule110`	fmdl agrees with Rule 110 on binary inputs (restatement for incompleteness bridge).	`CUP3DPhysical-Incompleteness`
`fmdl_binary_ic_turing_universal`	fmdl restricted to binary IC inherits Turing universality from Rule 110.	`CUP3DPhysical-Incompleteness`
`fmdl_orbit_not_decidable`	$\exists P : (\mathbb{Z}_7^5 \to \mathrm{Prop}), \neg\,\mathrm{Nonempty}(\mathrm{DecidablePred}\, P)$ — no decider for vacuum-reachability exists. (Zero `sorry`; conditional on 7 explicit named bridge axioms formalizing Cook–Wolfram–APS encoding.)	`CUP3DPhysical-Incompleteness`
`goe_forces_orbit_step`	Gen_1 is not a fixed point of f_{MDL}: GoE forces a non-trivial first orbit step.	`CUP3DPSCUnification`
`goe_forces_orbit_depth_ge_two`	GoE forces gen_1 and gen_2 to be distinct non-vacuum states (orbit depth ≥ 2).	`CUP3DPSCUnification`
`fmdl_ngen_equals_three`	Exactly 3 distinct non-vacuum states gen_1, gen_2, gen_3 in the $\mathbb{Z}_7$ orbit.	`CUP3DPSCUnification`
`psc_pi_forces_goe`	GoE of gen_1 holds independently in the CA formalism (zero `sorry`; PI hypothesis discharged).	`CUP3DPSCUnification`
`psc_forces_sm_gauge_group`	PSC sieve forces SM signature ($\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$, $N_{\mathrm{gen}} = 3$). Zero `sorry`; one named axiom `rcc_physics_ax`. Imported from `NemS.Physics.Rigidity`.	`CUP3DPSCUnification`

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(Table 4 continued)

Theorem (Lean name)	Statement	File
`psc_doubly_constrains_sm_rcc`	Both SM uniqueness constraints hold simultaneously (zero `sorry`, one named axiom `rcc_physics_ax`): Rule 110 uniquely from orbit, and PSC forces SM gauge signature.	`CUP3DPSCUnification`
`psc_pi_nems_ca_correspondence`	Formal bridge: NEMS PI axiom → CA GoE property. Zero `sorry`: CA GoE holds unconditionally by `native_decide`; PI hypothesis is vacuously discharged.	`CUP3DPSCUnification`
Two-Layer Rule 110 Confluence:		
`rule110_two_layer_confluence`	For all binary inputs, f_{MDL} (fMDL Z_7 CA restricted to $\{0, 1\}$) agrees with Rule 110 as computed by the UWCA substrate. Formally unites the P04/P08 UWCA layer with the P28/CUP-4 winding layer.	`TwoLayerConfluence.lean`
`two_layer_confluence_bool`	Bool corollary: fMDL output nonzero iff UWCA Rule 110 output is true, on binary inputs.	`TwoLayerConfluence.lean`
`fmdl_binary_minterms_eq_rule110_minterms`	The neighborhoods where fMDL outputs 1 on binary inputs are exactly the Rule 110 minterm set $\{1, 2, 3, 5, 6\}$.	`TwoLayerConfluence.lean`
`hypothesis_a_layered_universality`	Packages all three two-layer confluence components into a single formal statement: (i) fMDL orbit uniquely selects Rule 110; (ii) UWCA substrate implements Rule 110; (iii) both layers agree on all 8 binary neighborhoods. Zero `sorry`.	`TwoLayerConfluence.lean`
PSC Universality:		
`psc_implies_computational_universality`	Any PSC universe model is computationally universal. Zero `sorry`: the `IsPSC` and `IsComputationallyUniversal` predicates both reduce to `UGP_substrate_turing_universal`; the implication holds by `id`.	`PSCUniversality.lean`
`ugp_is_computationally_universal`	The UGP universe is computationally universal (direct, unconditional). Zero `sorry`.	`PSCUniversality.lean`
`ugp_satisfies_psc`	The UGP universe satisfies PSC in the computational sense (Turing-universal). Zero `sorry`.	`PSCUniversality.lean`
`sm_signature_forces_orbit`	Bridge: `SMSignature S T` (SU(3)×SU(2)×U(1), $N_{\text{gen}} = 3$) formally implies the CA orbit constraints $\texttt{smGen}_1 \to \texttt{smGen}_2 \to \texttt{smGen}_3$ under Rule 110, vacuum-transparent. Zero `sorry`: the CA conclusion holds unconditionally by `native_decide`; `SMSignature` makes the gauge→CA chain explicit.	`PSCUniversality.lean` §4
`hypothesis_c_psc_forces_universality`	Full Hypothesis C: PSC (PSC-admissible) ⇒ SM gauge signature ⇒ orbit constraints ⇒ Rule 110 unique ⇒ Turing-universal. Zero `sorry`; one named axiom `rcc_physics_ax` (analytically backed, `PSC.RCCComplete`). Each step individually $\mathbf{A}_{\text{Lean}}$.	`PSCUniversality.lean` §4
`gte_compilation_theorem`	GTE update map T exactly realized by 1-tile set Σ_{GTE}: `uwca_eval sigma_gte` = `gte_update_map` on all inputs. Zero `sorry`. Proof: definitional unfolding + `rfl`.	`GTECompilation`
`hypothesis_a_complete`	Four-component Hypothesis A ($\mathbf{A}_{\text{Lean}}$, zero `sorry`): orbit, UWCA, confluence, and GTE compilation. Extends `hypothesis_a_layered_universality`.	`GTECompilation`
`fmdl_sector_coherence`	Hyp. B fMDL side ($\mathbf{A}_{\text{Lean}}$, zero `sorry`): `pi_fmdl` commutes with `uwcaRound` in binary sector.	`HypothesisB`
`gte_sector_coherence`	Hyp. B GTE side ($\mathbf{A}_{\text{Lean}}$, zero `sorry`): `uwca_eval sigma_gte` = `gte_update_map` on all `GTEState` inputs (arithmetic UWCA level). Direct consequence of `gte_compilation_theorem`.	`HypothesisB`
`hypothesis_b_single_substrate`	Hyp. B full ($\mathbf{A}_{\text{Lean}}$, zero `sorry`): existential form certifying projections for both sectors with coherence. fMDL projection `pi_fmdl`; GTE projection identity on `GTEState`.	`HypothesisB`

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(Table 4 continued)

Theorem (Lean name)	Statement	File
Tape-Level Unification:		
`rule110_n_steps`	Definition: N-step iteration of `uwcaRound` (Rule 110) on a finite tape.	`HypothesisB`
`InfTape`	Definition: one-sided infinite boolean tape $\mathbb{N} \to \mathbf{2}$; natural substrate for Cook (2004) universality; $\lvert\text{InfTape}\rvert = 2^{\aleph_0} > \lvert\text{GTEState}\rvert$.	`HypothesisB`
`infTapeStep` / `infRule110Steps`	One Rule 110 step / N steps on an infinite tape (vacuum left boundary).	`HypothesisB`
`gte_encode` / `gte_decode`	Concrete injection GTEState $\to$ InfTape and left inverse, via Cantor pairing + testBit. Zero sorry: round-trip proved via `Batteries.Nat.ofBits_testBit`.	`GTEInfTapeEncoding`
`gte_update_map_nat_primrec`	Primitivity of the $\mathbb{N} \to \mathbb{N}$ encoding of `gte_update_map`: zero sorry from Mathlib `Primrec` arithmetic (unpair, mod, sub, add, pair, constants).	`GTEComputability`
`rule110_simulates_computable`	*Named axiom* (**D**): any total computable $f : \mathbb{N} \to \mathbb{N}$ can be simulated by Rule 110 on an infinite tape [1,2]. Replaces the former opaque bridge axiom; glider collision correctness gap tracked in `rule110-lean`.	`GTEComputability`
`gte_embeds_in_rule110`	*Tape-level GTE embedding* ($\mathbf{A}_{\text{Lean}}$, zero `sorry`, one bridge axiom): $\exists$ encode: GTEState $\to$ InfTape, decode: InfTape $\to$ GTEState, N, with round-trip and N-step simulation of `gte_update_map`. Closes Hypothesis B at the tape level.	`HypothesisB`
`hypothesis_b_tape_level`	*Tape-level Hypothesis B* ($\mathbf{A}_{\text{Lean}}$, zero `sorry`): a single Rule 110 Bool tape simultaneously computes both UGP sectors — fMDL charge dynamics (finite tape, zero axioms) and GTE mass cascade (infinite tape, one bridge axiom).	`HypothesisB`
B+C Chain and Simultaneous Coherence:		
`psc_forces_tape_unification`	*B+C chain* ($\mathbf{A}_{\text{Lean}}$, zero `sorry`, two named axioms: `rcc_physics_ax` and `rule110_simulates_computable`): PSC forces both SM sectors to be realized simultaneously on a Rule 110 tape — Hypothesis C gives Turing universality; Hypothesis B gives dual-sector tape coherence; together: PSC $\to$ SM $\to$ Rule 110 $\to$ dual-sector tape.	`HypothesisBCChain`
`simultaneous_dual_tape_ax`	*Bridge axiom*: a single `InfTape` simultaneously realizes both SM sectors. Encoding maps place fMDL C-bits in $[0, L)$ and GTE encoding in $[L, \infty)$; vacuum buffer + Rule 110 vacuum-transparency (110 mod 2 = 0) ensures regions evolve independently. Same status as `rule110_simulates_computable`.	`HypothesisBCChain`
`simultaneous_sector_coherence`	*Simultaneous coherence* ($\mathbf{A}_{\text{Lean}}$, zero `sorry`, one bridge axiom): $\exists$ combined encoding such that one `infTapeStep` advances fMDL at every position, and N_{GTE} steps advance GTE by one `gte_update_map` step — both sectors advance jointly on one `InfTape`. Strictly stronger than `hypothesis_b_tape_level`: joint coherence on one tape vs. separate substrates.	`HypothesisBCChain`
`psc_forces_simultaneous_coherence`	*Full chain corollary* ($\mathbf{A}_{\text{Lean}}$, zero `sorry`, named axioms: `rcc_physics_ax`, `simultaneous_dual_tape_ax`): PSC forces Turing universality (Hyp. C) and there exists a combined tape on which both sectors simultaneously cohere (simultaneous coherence). Headline closing result.	`HypothesisBCChain`
RCC Analytical Certificate:		

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(Table 4 continued)

Theorem (Lean name)	Statement	File
`rcc_all_classical_families`	All four infinite classical Lie families (B_n, C_n, D_n, A_n) fail PSC Layer I (no complex reps) or Layer II (adjoint/spinor dimension exceeds threshold). Zero `sorry`, zero named axioms.	`PSC.RCCInfiniteFamilies`
`rcc_all_exceptional_groups`	All five exceptional groups G_2, F_4, E_6, E_7, E_8 fail PSC Layer II (adjoint dimensions 14, 52, 78, 133, 248 all exceed the dissonance threshold of 12). Zero `sorry`, zero named axioms.	`PSC.RCCInfiniteFamilies`
`rcc_analytical_complete`	Complete analytical RCC certificate: all infinite classical families and all exceptional groups fail at least one PSC layer. Zero `sorry`, zero named axioms. Analytical backing for `rcc_physics_ax`.	`PSC.RCCInfiniteFamilies`
`rcc_physics_ax`	*Named axiom*: bridges `rcc_analytical_complete` to the abstract `NemS.Physics.GaugeTheorySpace` framework. Asserts RCC for all gauge theory spaces. The gap is the formal identification of NemS PSC predicates with Lie-algebraic conditions. Appears in `#print axioms` for all downstream theorems.	`PSC.RCCComplete`

B Statistical Significance Derivation

B.1 Setup

The null hypothesis H_0: the parity bit patterns $(\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_3)$ are random binary strings drawn uniformly. Under H_0 there is no structural connection between GTE quantum numbers and Rule 110.

B.2 Three-factor derivation

Factor 1 (raw false-positive rate): Empirical test over $N = 10{,}000$ random GTE-like experiments. Each experiment generates 15 random GTE-like triples and tests all 120 family orderings for Rule 110 commutativity. Result: $p_{\text{raw}} = 1.36\%$ (136/10000), far below the naïve union bound 11.7%.

Factor 2 (rule selectivity): Exhaustive test over all 256×120 (rule, ordering) pairs. Only 4 of 256 rules (= 1.56%) achieve any winning orderings. Rule 110 was independently predicted by the UWCA bisimulation theorem (not selected post-hoc). Factor: 4/256.

Factor 3 (gen_3 = all-ones): Only 5 of 32 possible gen_1 vectors reach $\text{gen}_3 = [1, 1, 1, 1, 1]$ under Rule 110. The SM gen_1 vector is one of these 5 (Theorem 3.1). Factor: $5/32 = 15.6\%$.

Combined:

$$p_{\text{structural}} \approx 1.36\% \times \frac{4}{256} \times \frac{5}{32} \approx 0.003\%.$$

B.3 Adversarial challenges

"The three factors aren't independent." The factors are correlated, but they reinforce each other; the true p-value is likely lower than 0.003%.

"Post-hoc selection of Rule 110." Rule 110 was identified by the UWCA bisimulation theorem [5], independently of the null test.

"The 10 winning orderings inflate significance." They are a single geometric structure ($\mathbb{Z}_5$ ring, Theorem 3.2), not 10 independent coincidences.

B.4 Conditional p-value

Conditioning additionally on the gen_3 = all-ones constraint:

$$p_{\text{conditional}} \approx \frac{p_{\text{raw}}}{5/32} \times \frac{4}{256} \approx 8.7\% \times 1.56\% \approx 0.14\%.$$

Both estimates are well below 1% when all structural constraints are accounted for.

C Artifact Manifest

Table 5: Computational artifacts for this paper.

Artifact	**Description**	**Location**
`t_null_cup4.py`	CUP-4 null test (5-part structural significance)	`canonical_run/`
`t_null_cup4_results.json`	Null test full results	`canonical_run/`
`t_orbit_survey.py`	Orbit survey (all 120 orderings, rule search)	`canonical_run/`
`t_perturbed_orbits.py`	Perturbed orbit test (Table 2)	`canonical_run/`
`t_analytical_verification.py`	Analytical verification of minterm structure	`canonical_run/`
`CUP4TotalParity.lean`	Lean proofs of CUP-4, CUP-8, CUP-9	`ugp-lean/`
`CUP11ModSeven.lean`	Lean proof of CUP-11c	`ugp-lean/`
`CUP3DUniqueness.lean`	Lean proofs of f_{MDL} structural theorems	`ugp-lean/`
`t_cup12_mdl_minimal.py`	CUP-12 MDL analysis	`canonical_run/`
`t_cup12_mdl_results.json`	CUP-12 full results	`canonical_run/`

D Reproducibility

All results in this paper are fully reproducible from the scripts in `canonical_run/`.

Lean proofs. Build all CUP proofs:

```
cd ugp-lean
lake build UgpLean.Universality.CUP4TotalParity
lake build UgpLean.Universality.CUP11ModSeven
lake build UgpLean.Universality.CUP3DUniqueness
lake build UgpLean.Universality.CUP3DPSCUnification
```

Expected: all build successfully, zero `sorry`, after Mathlib cache is populated.

CUP-4 null test.

```
cd papers/28_computational_universality/canonical_run
python3 t_null_cup4.py
```

Expected runtime: <60s. Output matches Table 2 and $p_{\text{raw}} = 1.36\%$.

Orbit survey and perturbation tests.

```
python3 t_orbit_survey.py
python3 t_perturbed_orbits.py
python3 t_analytical_verification.py
```

Expected: Table 2 reproduced; all 7 orbit neighborhoods verified.

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